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Leveraging AI to Enhance Knowledge Sharing and Innovation in Agile Project Management

A Study on How AI Can Support Agile Project Management and Foster Effective Knowledge Sharing

Ines Lindstedt Hidiri

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Ines Lindstedt Hidiri

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Date: May 22, 2025
Examiner: Sofia Ritzén
Academic adviser: Susanne Nilsson
Industrial adviser: Fredrik Carlsson & Eric Toste

KTH Royal Institute of Technology
School of Industrial Engineering and Management (ITM)
Department of Engineering Design
SE-100 44 Stockholm, Sweden

Abstract

This study explores the integration of artificial intelligence (AI) into agile project management to enhance knowledge sharing and innovation within project-based organizations. In an increasingly complex and fast-paced business environment, AI is identified as a strategic enabler that can reinforce agile practices and strengthen long-term innovation capabilities. The research combines a systematic literature review with a case study conducted at a leading energy company to identify key opportunities and challenges associated with AI implementation.

Findings indicate that AI can reduce administrative workloads, optimize resource allocation, and enable employees to focus on creative and value-adding activities. Successful integration of AI requires gradual implementation aligned with agile principles, continuous training, and a collaborative organizational culture. AI should be positioned as a support for agile methodologies rather than a replacement for human decision making.

The study also addresses specific considerations for organizations with strict information security requirements and presents a structured implementation framework. Overall, the results highlight a clear need for AI support in project processes, with the potential to significantly improve efficiency, knowledge exchange, and innovation outcomes.

Key words

Artificial intelligence, agile project management, knowledge sharing, innovation, AI implementation

Sammanfattning

Denna studie undersöker hur artificiell intelligens (AI) kan integreras i agil projektledning för att stärka kunskapsdelning och innovation inom projektbaserade organisationer. I en alltmer komplex och snabbrörlig affärsmiljö identifieras AI som ett strategiskt verktyg för att förstärka agila arbetssätt och långsiktigt stärka organisationers innovationsförmåga. Studien kombinerar en systematisk litteraturgenomgång med en fallstudie på ett ledande energiföretag för att identifiera centrala möjligheter och utmaningar vid AI-implementering.

Resultaten visar att AI kan minska administrativt arbete, optimera resursanvändning och möjliggöra för medarbetare att fokusera på kreativt och värdeskapande arbete. För en framgångsrik integration krävs en stegvis implementering i linje med agila principer, kontinuerlig utbildning samt en kultur som präglas av samarbete. AI bör ses som ett stöd för agila arbetssätt och inte som en ersättning för mänskligt beslutsfattande.

Studien belyser även särskilda hänsynstaganden för organisationer med höga krav på informationssäkerhet och presenterar ett strukturerat implementeringsförslag. Sammantaget pekar resultaten på ett tydligt behov av AI-stöd i projektprocesser, med potential att öka effektivitet, kunskapsutbyte och innovationskraft.

Nyckelord

Artificiell intelligens, agil projektledning, kunskapsdelning, innovation, AI-implementering

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Acronyms

AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
LLM	Large Language Model
GPT	Generative Pretrained Transformer
BERT	Bidirectional Encoder Representations from Transformers
CI	Collective Intelligence
ICT	Information and Communication Technology
PBO	Project Based Organizations
API	Application Programming Interface
AIMS	AI Management Systems
XAI	Explainable AI
UX	User Experience
UI	User Interface

1. Introduction

The following chapter provides an introduction to the study with the objective to provide context and understanding of the research topic and the limitations of the study.

1.1 Background

In today's fast paced and uncertain business landscape, organizations face increasing pressure to shorten time-to-market, meet evolving customer demands, and deliver continuous value (Balasubramaniam et al., 2022). At the same time, product development environments are often characterized by fragmented knowledge, lack of transparency, inefficient collaboration, and challenges in decision making, particularly in large, distributed, or siloed organizations. These issues can slow down innovation and undermine long-term competitiveness (Gieskes et al., 2002).

To address these challenges, there is growing interest in leveraging Artificial Intelligence (AI) to streamline internal processes and drive innovation (Haefner et al., 2021). However, the practical application of AI in organizational settings remains relatively limited, especially in agile, innovation-driven project environments (Cabrero-Daniel, 2023). Moreover, there is a lack of empirical research and established frameworks that explore how AI can be meaningfully and strategically integrated into existing project environments. This reveals a growing need to move beyond theoretical discussions and investigate how AI can support real-world agile and innovative ways of working.

Introducing AI into project workflows also presents several challenges. Issues such as data quality, algorithmic bias, lack of transparency, and the "black box" nature of many AI models can complicate decision-making and accountability (Fairfax, 2024; Jin, 2024). Over reliance on AI may reduce human judgment and creativity, both of which are fundamental to agile methodologies. Furthermore, ethical considerations, data privacy regulations, and high implementation costs add further complexity to AI adoption (Taylor, 2022; IBM, 2021).

Agile project methodologies emerged in the mid-1990s in response to the need for increased flexibility, customer focus, and adaptability to changing requirements (Zaveri et al., 2024). These methodologies have since transformed how projects are managed by emphasizing iterative development, continuous feedback, and cross-functional collaboration. However, as projects have become more complex, there is a growing need to further develop and optimize agile approaches (Kaur et al., 2015).

In this context, AI has been identified as a potentially powerful enabler of agile evolution. AI technologies can support teams by identifying patterns, predicting risks, automating routine tasks, enabling data driven decision-making, and improving knowledge sharing capabilities that directly address many existing bottlenecks in innovation work. Several recent studies (e.g., Jin, 2024; IBM, 2021; Fairfax, 2024) highlight AI's potential to improve planning, resource allocation, and decision-making in project environments.

However, research on the integration of AI into agile projects remains in its early stages (Cabrero-Daniel, 2023). Few studies have examined its practical impact or long-term consequences, and much of the existing literature is based on theoretical assumptions rather than empirical evidence. This creates a gap between the perceived potential of AI and the concrete strategies required for its successful implementation in innovation-driven environments. In particular, there is a need to better understand how AI can be used to support core agile principles such as collaboration, adaptability, and continuous learning (Elham Karim Zadeh et al., 2024). Since effective knowledge retention and sharing are crucial for driving innovation (Fairfax, 2024), this study focuses on exploring how AI can strengthen these aspects within agile project management.

1.2 Purpose and Research Questions

The purpose of this thesis is to investigate how Artificial Intelligence can be strategically integrated into agile project management practices at Vattenfall's R&D department to enhance knowledge sharing, improve knowledge retention and accessibility, and increase operational efficiency. By examining current practices and identifying opportunities for AI integration, the study aims to contribute to more effective knowledge management and ultimately foster innovation in agile teams.

- RQ1: What applications of AI can facilitate knowledge retention and accessibility in an agile environment?
- RQ2: How can AI be integrated to support agile project management to strengthen knowledge sharing as a foundation for innovation?

1.3 Delimitations

The time frame for this study was limited to 20 weeks, which frames the scope and the collection of data. Additionally, the study was confined to Vattenfall and projects within the Research & Development department. Furthermore, the research does not explore the technical development of AI models but instead focuses on their strategic and managerial applications. The study primarily relies on qualitative data, gathered through interviews and document analysis. Due to these constraints, the findings may not be fully generalizable to all industries or companies but aim to provide valuable insights into AI's role in agile project management within similar organizational contexts.

2. Literature

This chapter provides an overview of the current literature on the use of AI in agile project management. The focus is on how AI can be integrated into agile processes, as well as the challenges and benefits that may arise when implementing AI technologies.

2.1 The Evolution and Application of Agile Methodology

During the last decade the agile model has become the dominant approach in project management (Kaur et al., 2015). Its popularity stems from key features such as the ability to accommodate change requests at any stage of the project, a strong focus on client satisfaction, iterative development cycles, and continuous collaboration between clients and developers (Bohdan Haidabrus, 2024) (Kaur et al., 2015). The agile methodology originated in software development and is based on the principles outlined in the Agile Manifesto from 2001 (Cunningham, 2019), stated as:

- Individuals and interactions over process and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan.

Over time, the agile methodology has evolved beyond its original domain and is now applied in a wide range of project contexts, driven by its proven ability to deliver high-quality, functional, and innovative results (Annosi et al., 2024). Agile serves as a project management framework that structures work into iterative phases, often organized as sprints, making it particularly effective for smaller projects, as it allows for flexibility in managing scope and requirements lifecycle (Kaur et al., 2015). Two of the most widely used agile approaches, Scrum and Kanban, offer distinct methods to enhance efficiency and adaptability in project execution (Hohl et al., 2018). Agile methodologies emphasize innovation and the ability to quickly adapt to changing customer needs and environmental factors, a key characteristic that allows agile teams to remain flexible and responsive in dynamic settings (Annosi et al., 2024).

2.2 AI in Agile Project Management

AI has emerged as a key technology with the potential to revolutionize industries (Haefner et al., 2021). The integration of AI into agile environments is stated to mark the beginning of a new era in project management, characterized by enhanced efficiency, improved planning accuracy, advanced decision-making capabilities, and more effective knowledge sharing. As agile methodologies continue to evolve, AI integration will become increasingly essential for fostering innovation and ensuring successful project outcomes (Fairfax, 2024).

When implementing AI in agile environments, it is essential to understand the core principles of agile methodologies, such as collaborative teamwork, adaptive planning, and early delivery (Kaur et al., 2015) - as well as to have a solid understanding of AI itself. The agile methodologies emphasizes the importance of direct communication and collaboration, prioritizing them over strict compliance with formal processes. At the same time AI harnesses algorithms and data to enhance human capabilities, automate tasks and make predictive insights. When these two fields converge, AI may enhance the agile process by processing large datasets, predicting potential project risks, and improving resource management. This potentially leads to a more adaptive and data-driven agile framework that evolves dynamically in real time (Fairfax, 2024).

Planning

Planning is a crucial stage in agile projects and the traditional planning approaches often depend on human judgment and past data, which can be limited or influenced by bias (Fairfax, 2024). In contrast, AI enhances the planning process by enabling project managers to make more accurate forecasts and track progress more effectively, reducing the impact of biases and improving overall project outcome (Jakkula et al., 2023) (Jin, 2024). By predicting timelines, resource needs, and potential bottlenecks, AI helps create a realistic schedule and prioritize tasks to keep the project on track. Additionally, it provides valuable insights into the ongoing project's progress (Zahaib Nabeel, 2024). Using historical project data, as well as team member skills, AI can suggest schedule adjustments, optimize resource allocation, and set task priorities (Polonevych et al., 2020) (Jin, 2024). Becoming a valuable knowledge-sharing tool within project planning. This saves the project manager significant time and provides the team with real-time progress updates. This will hopefully ensure that the projects complete at set time (Jakkula et al., 2023).

Additionally, the project manager can focus more on collaboration and innovation rather than time-consuming planning tasks (Polonevych et al., 2020). If implemented across the company, this AI tool can standardize the planning approach, making it easier for employees to adapt to the project structure. It also fosters consistency across teams and helps estimate time for different tasks based on previous projects. And by focusing on measurable outcomes like cost efficiency and on-time delivery, the authors effectively communicate their findings and provide actionable insights for managers (Fairfax, 2024).

Predictive analysis

Using AI powered predictive analytics can enhance the risk management through a more effective risk modeling, as well as forecast project risks and delays with remarkable accuracy, which enables the project managers to prevent and address the risks before they escalate (Bogojevic Arsic, 2021). However, AI can also be used to predict market trends and potential operational bottlenecks (Jakkula et al., 2023).

AI can leverage data from previous projects to generate more accurate cost estimates and proactively manage risks, enabling project managers to better prepare for various scenarios. This approach supports more accurate budgeting and resource allocation (Jakkula et al., 2023). Additionally, AI can provide live cost tracking and suggest cost-effective alternatives for resources, suppliers, or processes (Vandana M. & Dr. T. Lavanya Kumari, 2024). Upon project completion, AI can analyze budget data, highlight patterns, and enhance cost efficiency for future projects (Zahaib Nabeel, 2024).

This ability to optimize future performance is further supported by a study from Gartner (2020), which found that predictive analytics can reduce project delays by up to 30% by identifying and mitigating risks early. Similarly, McKinsey & Company (2020) discovered that organizations utilizing AI for predictive insights experienced a 20-30% improvement in project efficiency. These insights help agile teams by providing foresight into potential bottlenecks, risk factors, and areas requiring resource optimization (Fairfax, 2024).

Automating routine tasks

Every agile team faces a multitude of repetitive and time consuming tasks. One key benefit of AI integration is the automation of these tasks (Jin, 2024). AI can assist with things as code review and testing, and can leverage this with high precision and speed. This reduces the manual workload and allows the team to focus more on strategic activities such as innovation and problem-solving, as well as it helps with identifying patterns and make intelligent decisions and also reduces the human error (Fairfax, 2024).

Building on this, Taylor (2022) discusses research indicating that by 2030, AI will handle 80% of project management tasks-including data collection, reporting, and predictive analysis-in a consistent and efficient manner.

Enhancing Decision-Making

Agile focuses on collaborative decision-making and continuous improvements. AI can enhance this by offering insights from both historical and real-time data analysis, enabling teams to make more informed decisions and drive ongoing improvements. It can analyze large and complex datasets to identify patterns and relationships that might not be identified by human analysts. This data-driven decision-making process helps teams adapt more effectively to changes during the project (Fairfax, 2024). It also ensures well-informed decisions by considering all relevant factors, reducing the influence of the project manager's emotions and personal biases (Jakkula et al., 2023).

Improving Communication and Collaboration

Agile methodologies thrive on strong communication and collaboration in the team, for example, by using AI, it can help with streamline interactions by summarizing meetings, tracking tasks, and even identifying potential communication issues. AI can also offer real-time feedback and recommendations to ensure that all the team members stay aligned and up to date during the projects (Fairfax, 2024; Jin, 2024).

Below in *Table 1*, is a table extracted from the report *Generative AI in Agile, Project, and Delivery Management* by Bohdan Haidabrus (2024). The table presents a comparison of agile frameworks with and without generative AI integration. A closer look at the table clearly highlights the differences between traditional agile methods and an AI-integrated approach. It shows how AI can help teams accelerate their work by reducing time spent on repetitive tasks, allowing them to focus more on innovation and creative problem-solving.

Table 1: Comparison of agile frameworks with and without AI integration.

Criteria	**Tradition Agile**	**Generative AI Agile**
Efficiency and Speed	Agile relies on human resources for every aspect.	Can automate routine tasks, significantly speeding up the development process.
Decision Making and Planning	Relies on team expertise for decision-making.	Can provide predictive insights for better planning.
Scope of Innovation	Limited by the team's collective knowledge.	Can suggest innovative solutions beyond the team's knowledge base.
Risk Management	Manages risks based on experience, tools, and technics.	Can predict potential risks using data analysis.
Personalization and User Focus	Both methodologies focus on user needs, but Agile with Generative AI can analyze user data for deeper insights.	
Learning and Adaptation	Relies on feedback loops for learning.	Can facilitate rapid learning through data analysis.
Communication and Collaboration	Emphasizes human interaction.	Can automate communication while potentially impacting team dynamics.
Skill Requirements	Requires a range of technical and soft skills.	Demands understanding AI tools and data literacy.
Quality and Reliability	Both approaches focus on quality, but Agile with Generative AI can introduce new types of errors requiring human oversight.	
Ethical Considerations	Focuses on product and team ethics.	Adds considerations around data privacy and AI bias.

2.3 AI Models

To better understand AI's role, it's useful to look at its core components. What defines AI is its ability to simulate human cognitive functions. Some of its core components include ML, Deep Learning, Natural Language Processing (NLP), Computer Vision, and Robotics (Fairfax, 2024). In *Figure 1*, a visual representation of the different AI models and their interconnections is presented.

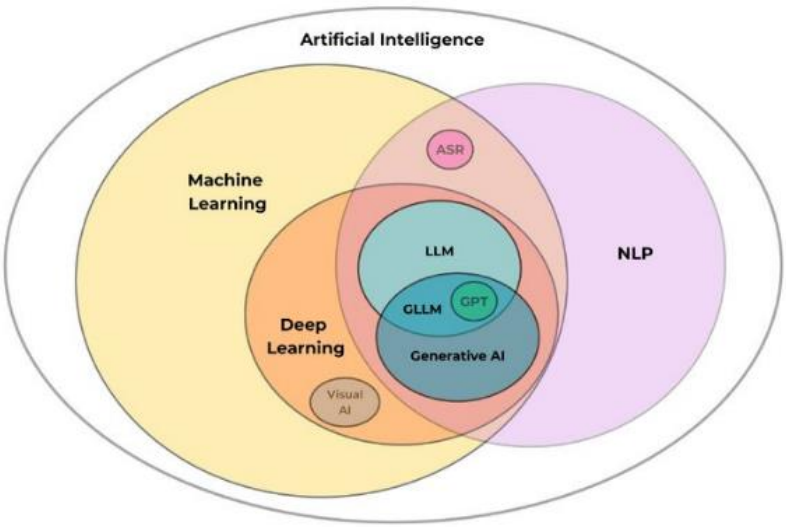


Figure 1: Visual representation of AI models.

ML is a branch of AI focused on developing algorithms that enable computers to learn from data and make predictions. It includes key areas such as supervised, unsupervised, and reinforcement learning. By analyzing patterns and adapting over time, ML allows systems to enhance their performance without requiring explicit programming for each task. A more specific area within ML is Deep Learning which uses neural networks with multiple layers to model complex patterns in data, this is most used in areas like image and speech recognition (Brown et al., 2020; Fairfax, 2024).

Another vital part of AI is Natural Language Processing, which enables computers to interact with humans using natural language. NLP is used in areas like speech recognition, sentiment analysis and translation (Brown et al., 2020). An advanced application of this is the Large Language Model (LLM), a type of NLP model trained on vast amounts of text data to generate and understand human language in a contextual and coherent manner. These models utilize neural networks with billions of parameters, trained through a self-supervised learning approach on massive datasets. By pre-training on extensive corpora from the web, LLMs can recognize complex patterns, capture linguistic nuances, and establish deep semantic connections. For example is GPT (Generative Pretrained Transformer) trained by internet text data and BERT (Bidirectional Encoder Representations from Transformers) trained by BookCorpus and Wikipedia (Azam et al., 2024).

This allows AI to comprehend context and logic, producing meaningful and relevant responses. It leverages deep learning techniques, particularly transformer architecture, to generate text, answer questions, write code, and more (Atkinson-Abutridy, 2024). Advances in this field have led to the development of models like GPT-3 and GPT-4, which is extremely powerful and can generate human-like text and understand context with impressive accuracy, for example ChatGPT (Brown et al., 2020; Atkinson-Abutridy, 2024).

To build a powerful system, knowledge fusion of LLMs is a common approach. An advanced method of this is FUSELLM, which externalizes knowledge from multiple LLMs and transfers their capabilities to a target model. This enables a system to leverage the expertise of different models, resulting in a more robust and adaptable AI solution. In some cases, integrating other tools alongside LLMs can further enhance performance and unlock the system's full potential (Wan et al., 2024). For example, Sentence-BERT can be used for semantic search, while ElasticSearch, a search engine, can optimize the retrieval process, improving efficiency. By combining these models, a more effective and scalable solution can be achieved (Wan et al., 2024).

2.3.1 Solutions on the Market

There are several AI-driven solutions available on the market that can help improve collaboration, streamline communication, and enhance productivity in agile environments. For instance, collaboration platforms like Microsoft Teams and Slack offer integrated AI functionalities to help teams maintain efficient interaction and keep up with the agile cadence. These tools provide features such as real time messaging, document sharing, and task tracking, all powered by AI to improve team dynamics and decision making (Fairfax, 2024).

Integrating AI technology, platforms such as monday.com, Writer.com, NotebookLM, and Microsoft Copilot offer solutions that enhance workflow efficiency, content creation, information management, and overall productivity. Other notable AI-driven platforms include ChatGPT, Google Assistant, and Claude AI, each providing unique features tailored to various user needs. Vattenfall use Microsoft as their main communication platform and some employees at Vattenfall have the license to use Microsoft Copilot, it can answer questions, edit documents, summarize information, and even respond to emails, making it a valuable resource for employees.

2.3.2 Concerns and Challenges with AI

While the integration of AI into project management offers substantial benefits, it also presents several critical challenges that organizations must carefully address, including data privacy concerns, integration complexity, and the need for continuous training and access to large, high-quality datasets (Taylor, 2022; Jin, 2024). AI can provide advantages such as early fault detection, but success hinges on having high-quality, sufficient data, as low-quality data reduces accuracy and relevance (Shahin et al., 2023). Transparency and explainability also pose issues, as AI systems are often seen as “black boxes,” which can hinder trust. Techniques like Explainable AI (XAI) help clarify decision-making and improve understanding (Jin, 2024). Other concerns include algorithmic bias, loss of human insight, and data privacy (Singh et al., 2023; Jin, 2024). Additionally, the ISO/IEC 42001:2023 standard outlines requirements for responsible AI use, aiming to balance innovation with governance, which may limit how freely AI can be applied within organizations (ISO, 2023). These factors require a holistic approach, considering technical, ethical, and cultural factors. Human judgment must remain central, with AI acting as a supportive tool rather than a replacement (Fairfax, 2024).

2.4 Knowledge Management in Project Based Organizations

One of the biggest challenges in agile project management is ensuring effective knowledge sharing, which is essential for project success and the development of high collective intelligence (CI) (Cui & Yasseri, 2024). As a key resource for value creation, well-structured knowledge exchange fosters both creativity and innovation within organizations (Luo & Xiao, 2024; Abbas & Felix, 2024). Information and communication technology (ICT) plays a crucial role in facilitating knowledge sharing by reducing barriers, improving data access, and streamlining related processes (Phung et al., 2016).

However, despite its importance, knowledge sharing can still be hindered by several barriers, both individual and organizational, as well as technological limitations. Therefore, it is essential to identify and understand these challenges. On an individual level, employees may fear appearing incompetent, lack motivation, perceive time constraints, or struggle to communicate their knowledge effectively. Organizational barriers include siloed structures, a lack of incentives, weak leadership support, and high employee turnover (Phung et al., 2016). Siloed structures, or "silos," refer to isolated departments or teams that operate with limited communication and collaboration with other parts of the organization. This fragmentation makes it difficult to access relevant information, share insights across projects, and build on previous knowledge, ultimately slowing down innovation and reducing efficiency (Bento et al., 2020). Technological obstacles may involve complex or scattered system, poor integration between tools, an unintuitive user experience, or that it's hard to locate (Phung et al., 2016).

In project-based organizations (PBOs), where work is carried out by temporary teams, knowledge sharing is especially challenging. The focus on short-term project goals often overshadows efforts to capture and transfer knowledge across projects, making long-term knowledge retention difficult (C.C. van Waveren et al., 2014). Although knowledge transfer is well researched, few empirical studies focus on its implementation in PBOs. While project management methodologies may address knowledge capture and transfer, these activities are rarely integrated into the overall project plan. (C.C. van Waveren et al., 2014).

In contrast, agile methodologies, with their emphasis on interactions, adaptability, and flexibility, promote strong collaboration within teams (Dingsøyr et al., 2012). However, this focus on team-specific, tactical knowledge, which often resides with individuals, can lead to knowledge loss if not properly managed (Santos et al., 2014). To mitigate this, organizations need processes to preserve and share knowledge effectively (Amritesh & Misra, 2014). Although agile methodologies encourage knowledge sharing within teams, their informal and adaptive nature can make it difficult to transfer knowledge across the organization (Hansen et al., 1999). The lack of formal practices outside the team can hinder long-term knowledge retention. Agile approaches tend towards personalization, focusing on informal knowledge flow, in contrast to traditional methods that favor codification (Hansen et al., 1999). Effective knowledge sharing requires adopting specific practices and providing organizational support (Santos et al., 2014).

2.4.1 Leveraging the SECI Model for Enhanced Knowledge Creation

To support continuous innovation and learning within organizations, the SECI model offers a valuable framework for enhancing knowledge creation. Nonaka emphasizes the importance of fostering a culture of learning, collaboration, and innovation (Rakan Aldmour, 2021). He distinguishes between two types of knowledge: tacit knowledge, which is personal, experience-based, and difficult to articulate, and explicit knowledge, which is structured, documented, and easily transferable (Adesina & Ocholla, 2019). Organizations that can effectively create, share, and utilize knowledge gain a strategic advantage in the market (Bertrand Moingeon & Edmondson, 1996).

According to Nonaka, knowledge management emerges through a dynamic interaction between tacit and explicit knowledge (Rakan Aldmour, 2021). This process is illustrated in Nonaka's SECI model, which consists of four stages: Socialization, Externalization, Combination, and Internalization (Rakan Aldmour, 2021). These stages describe how knowledge is continuously transformed and expanded within an organization (Adesina & Ocholla, 2019).

By integrating the SECI model, agile teams can more effectively manage and leverage knowledge throughout the project lifecycle. AI plays a pivotal role in supporting each phase of this process, optimizing knowledge exchange, ensuring more efficient documentation, and facilitating continuous learning, thereby enhancing team collaboration and overall project performance (Böhm & Durst, 2024). The strength of the SECI model lies in its ability to connect knowledge creation with knowledge sharing, through its dynamic process that moves between tacit knowledge and explicit knowledge. This led to the formulation of the first research question*: What applications of AI can facilitate knowledge retention and accessibility in an agile environment?*. To explore this, Chapter 5.4 discusses the SECI model in relation to AI and agile project management.

2.5 Integrating AI in Agile Project Management

Integrating AI tools into existing agile workflows can be complex, as it often needs some changes into the current processes and tools. Therefore it is crucial to ensure that the AI systems are compatible with agile project management tools and frameworks, and that new data is continuously added to ensure a smooth and efficient operation. These integration tasks can be time consuming and might need a lot of resources, as well as planning, to achieve a seamless transition (Fairfax, 2024).

To successfully identify the needs and feasibility of an AI tool in agile project management, it is essential to first explore opportunities and establish clear requirements. The process should begin with a thorough needs assessment, followed by an analysis of existing processes and data to determine areas where AI could add value (Fairfax, 2024). Developing AI use cases can provide insights into its practical applications and help define its role within the organization. Once potential applications have been identified, assessing technical feasibility and setting clear objectives and key performance indicators (KPIs) will be crucial to ensure alignment with business goals.

Building on this foundation, it becomes equally important to evaluate the financial viability of the proposed AI solutions. Assessing Return on Investment (ROI) allows organizations to weigh the benefits of AI tools against their associated costs and implementation efforts (Fairfax, 2024). Setting clear objectives ensures progress aligns with expectations. This includes tracking performance metrics such as time saved on administrative tasks, improved decision-making, and how AI fosters innovation and collaboration within teams. Defining these metrics early in the process enables a structured evaluation of success and supports continuous improvement throughout the implementation journey (Wright, 2015).

Additionally, scalability and sustainability must be considered to ensure the AI tool can adapt to growing demands without compromising performance. This includes both the ability to expand the AI system and handle increasing amounts of data and iterations, as well as ensuring that users continue documenting in a way that keeps the AI's responses relevant and up to date. Implementing such a solution will require cross functional collaboration, bringing together expertise from different departments to support development and integration (Fairfax, 2024). To ensure AI is accessible and user friendly, it must be compatible with existing tools and workflows. This can be accomplished using an application programming interface (API), which facilitates seamless data exchange between IT systems (Samola, 2025).

Fairfax (2024) is frequently referenced in this study as one of the few researchers explicitly linking AI to agile methodologies. In his book *AI in Agile Projects: The Future of Project Management*, he builds on previous research and presents multiple case studies that outline both expected benefits and potential challenges of AI integration. However, these studies have not been systematically evaluated or followed up over time, reflecting a broader issue in the field - while research identifies AI's potential effects and risks, it lacks concrete strategies for implementation and risk mitigation. This highlights the need for further investigation into AI's practical applications and real world impact within agile projects.

2.5.1 Implementation Challenges

Implementing AI requires changes not only in technology and processes but also in employee roles, responsibilities, and the broader organizational culture and structure (Fairfax, 2024; Furxhi, 2021). These shifts can trigger resistance, often due to a lack of trust in leadership (Jin, 2024), emotional reactions to disrupted routines, or fear of lacking the skills needed for the new environment (Furxhi, 2021). To address this, managers must actively support employees by acknowledging their concerns, clearly communicating the reasons for change, and offering training to build competence and confidence (Furxhi, 2021). Poor communication, if not addressed, can leave employees feeling excluded and unsure about both the process and outcomes of AI integration.

Additionally, unrealistic timelines can hinder successful AI adoption. While urgency can drive progress, employees need sufficient time to understand and adapt to changes. Rushed implementation can undermine trust and breed skepticism about AI's effectiveness (Furxhi, 2021). A more effective approach is to start with a smaller scale, such as launching a proof of concept to demonstrate AI's value and reduce skepticism (Hur AI Förändrar Ledarskapsroller, 2025). This gradual rollout helps build trust and creates momentum for broader adoption. Furthermore, fostering a culture of curiosity, collaboration, and continuous learning is crucial. By introducing AI as a supportive tool and gradually integrating it into workflows, employees can experience its benefits firsthand. In this context, soft skills - such as empathy and communication - become crucial. They help create an environment in which AI is viewed as a helpful enabler rather than a disruptive threat (Hur AI Förändrar Ledarskapsroller, 2025).

2.5.2 Enhancing AI Adoption Through Implementation Mapping and Design

To further enhance AI adoption within agile project management, a structured approach that aligns AI capabilities with organizational goals is necessary. One effective method is Implementation Mapping, a framework that helps identify key implementation strategies, define actions, and ensure alignment with business objectives (Kang et al., 2024). By systematically addressing challenges such as AI adoption, user training, and integration into existing workflows, Implementation Mapping facilitates a smoother transition while maximizing AI's impact. This ensures that AI adoption leads to tangible improvements in efficiency, decision-making, and innovation within agile environments (Kang et al., 2024).
However, for employees to fully embrace a new AI solution, it must also be intuitive and easy to use. A key way to achieve this is through Nudging, which encourages employees to engage with the tool in the desired way. Additionally, strong User Experience (UX) and User Interface (UI) design can significantly enhance usability and user engagement (Ugochukwu Francis Ikwuanusi et al., 2024).

These subtle design choices increase the likelihood that employees adopt best practices, ensuring proper documentation and structured knowledge sharing. This not only strengthens the company's ability to foster innovation and long term learning (Lindström et al., 2019) but also lays the groundwork for AI-driven advancements. As documentation becomes more structured and interconnected over time, AI can leverage this wealth of data to draw meaningful conclusions and support more advanced use cases (Mohammed et al., 2023).

2.6 Summary

Given these challenges, implementing an AI tool for specific purposes can be difficult, as it relies on clear documentation and a consistent structure across the department. The lack of standardized documentation not only complicates AI implementation but also limits knowledge sharing and efficiency in project work. This gap highlights the need to refine project documentation guidelines to strike a balance between maintaining useful records and avoiding excessive documentation that contradicts agile principles.

Existing literature acknowledges the tension between documentation requirements and agile methodologies but lacks concrete strategies on how AI can support effective knowledge management without adding unnecessary administrative burdens. By addressing this gap, the study will provide insights into how AI can facilitate knowledge retention and accessibility in an agile environment. Specifically, it seeks to answer the following research questions:

- RQ1: What applications of AI can facilitate knowledge retention and accessibility in an agile environment?
- RQ2: How can AI be integrated to support agile project management to strengthen knowledge sharing as a foundation for innovation?

Below, in *Table 2*, is a summary of the key takeaways from the literature section.

Table 2: Summary of key takeaways from the literature section.

AI's potential in agile project management	• Enhanced Efficiency & Planning • Improved Forecasting • Predictive Analytics • Automation of Routine Tasks • Data-Driven Decision-Making • Handling large data volumes • Streamlined Collaboration & Communication • Supports knowledge sharing
Concerns and Challenges with AI	• Data Quality: • Transparency & Explainability: • Ethical Concerns: • Balancing AI & Human Involvement • International standards and Regulations • Limited empirical studies, relying on assumptions
Implementation Challenges	• Need of Organizational Change • Resistance to Change • Requires continuous testing
How to deal with implementation challenges	• Proof of Concept: • Fostering a Culture • Human soft skills and buildigh stust • Structured Approach • KPI's • Ensure scalability and sustainability
Enhancing AI Adoption	• Nudging & UX/UI Design • Increase user engagement by design choices

3. Methodology

The data was collected from various sources, including interviews, observations, internal documents, and informal meetings, in order to gather comprehensive insights. The methodology for data collection is presented below.

3.1 Case Description

This research project is conducted in collaboration with Vattenfall, a state-owned energy company with approximately 20,000 employees. Vattenfall operates across multiple European markets, including Sweden, Germany, the Netherlands, Denmark, and the United Kingdom, producing energy from diverse sources such as wind, biomass, hydro, gas, nuclear, solar, and district heating.

This thesis presents a case study within Vattenfall's Research and Development department (R&D), which operates as a project-based organization organized into portfolios and sections. The department consists of around 140 employees, divided into 7 portfolios across 9 sections. What distinguishes R&D at Vattenfall is their focus on developing solutions driven by both market pull and technology push. *Figure 2* presents the seven portfolios.

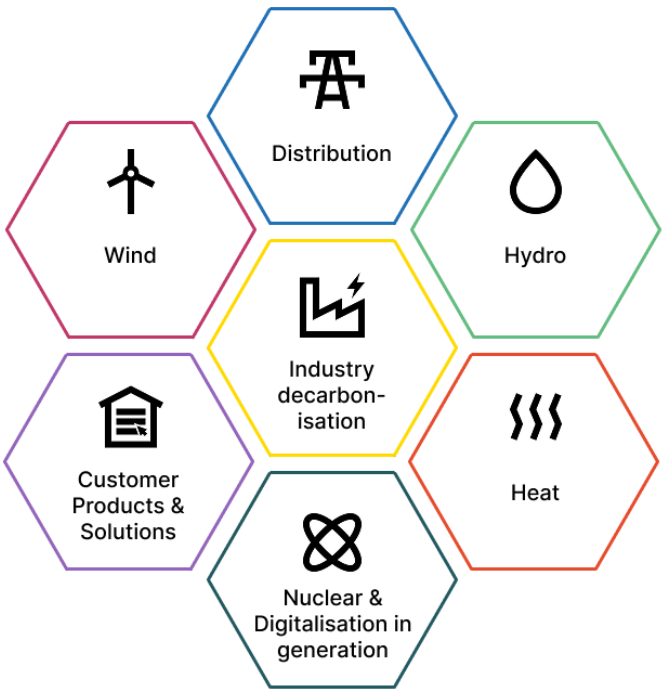


Figure 2: Vattenfall R&D, 7 portfolios at R&D.

From internal documents, some information about the project process was collected. There are two common scenarios for new projects within the R&D department. In the first scenario, internal team members generate ideas based on earlier studies or ongoing work, identifying areas that require further exploration or development. In the second scenario, project requests come from other departments within Vattenfall, where external teams present specific needs or challenges that the R&D department is tasked with addressing. The employees often have between 1-3 projects consisting of 1-4 members running at the same time, and are referred to as small projects. The timeline, number of projects, and team members are varying but normally around 1 year with 30% pace. In *Figure 3*, the organizational structure is visually represented. Each portfolio is managed by a Portfolio Manager, who oversees the Product Owners. The Product Owners, in turn, are responsible for specific project teams, each consisting of various Project Members.

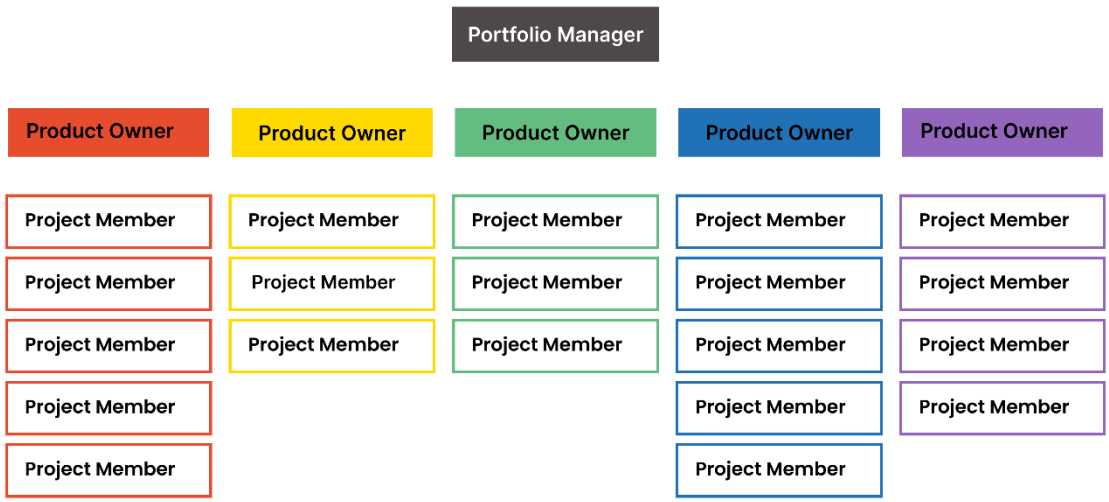


Figure 3: Organizational structure at R&D.

The above provides a brief introduction to Vattenfall R&D, but what mainly differentiates the organization from others are its strict confidentiality requirements. Information cannot be freely shared, which applies not only across different departments within the company but also within project teams.

3.1.1 Project Management Framework at Vattenfall

The employees at Vattenfall have a shared portal, in this report it is called *VF Portal*. In the portal there is information about roles, responsibilities, and mandatory documentation during projects. There are also some supporting documents, during the projects. Shortly the *VF Portal* can be described as a tool for starting up projects and taking them to the next level, as well as continuous progress reporting.

At the outset of a project, the team documents key information such as the project scope, timeline, and estimated duration. These estimates are based on factors including the project description, budget, team size, and a brief pre-study, though other elements may also influence them. The team also decides whether to adopt an agile or linear approach, ensuring that the chosen methodology aligns with the project's objectives and requirements.

Throughout the project, the team is responsible for providing regular progress updates, maintaining clear and consistent communication with relevant stakeholders to ensure alignment and transparency, and making decisions on whether the project should proceed at each tollgate (TG).

After the project is completed, the team typically delivers a report and a PowerPoint presentation, and in some cases, hands over the project to the client. Once this is done, the team should formally close the project and proceed with archiving. During the archiving process, any confidential information is removed from the project folder. Finally, the project is published in a shared folder, making it accessible to everyone within the portfolio. In *Figure 4* the project framework, *VF Portal*, with the mandatory documentation documents is presented.

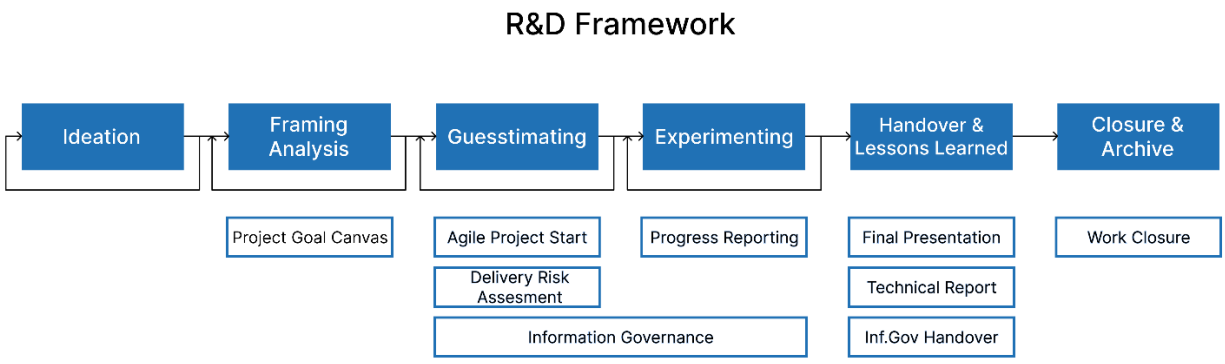


Figure 4: R&D project framework with mandatory documentation.

Besides all the documentation, they also conduct regular reviews. Usually they have project meetings with the project group 1-2 times a week, steering group meeting each month, product owner have follow-up meeting with the portfolio manager once a month and the project group have demo once a quarter (sometimes more often) where they present the projects for a general group at R&D, sometimes its completed project and sometimes not.

Vattenfall's documentation system is not used as intended. Interviews and observations show that while projects are successfully completed, many deviate from the established documentation plan, and some teams fail to properly archive their work. The exact extent of these administrative shortcomings remains unclear. Archived projects are stored in a shared R&D folder, while unarchived ones remain in project group cloud folders, increasing the risk of information loss and making it difficult to track past projects, conduct follow-ups, and share knowledge.

3.2 Research Design

The aim of this study is to explore how AI can be utilized in agile project management to enhance knowledge sharing among teams and team members, ultimately fostering greater innovation. The goal is to provide recommendations on how AI can be effectively applied within the R&D department at Vattenfall.

To formulate these recommendations and identify the most suitable solution for the company, this thesis adopts a comprehensive approach. The research follows a case study methodology, a qualitative research approach that is particularly effective for exploring specific contexts and gaining an in-depth understanding of particular issues and situations. An iterative approach was chosen to allow for the continuous integration of new insights as the analysis is based on a case company (Pär Blomkvist & Hallin, 2015).

The methodology was designed to support the research objectives, encompassing research design, data collection, and analysis. It combined a case study, qualitative literature review, and qualitative interviews. These qualitative methods facilitated a deeper exploration of the subject's nuances and challenges, while also enabling a comparison between theoretical concepts and real-world scenarios (Bryman & Bell, 2017).

3.3 Research Process

The research process began with defining a relevant research area and its delimitations. Consideration was given to both the requirements of KTH, Royal Institute of Technology and the needs of the case company, Vattenfall. This was carefully evaluated to ensure the delivery of results that aligned with the objectives of both parties. Based on this, research questions and objectives were formulated to be achieved throughout the project.

Next, the literature study was initiated, forming the foundation of the thesis and providing insights into existing research, methodologies, and theoretical frameworks. The literature was continuously updated throughout the project to ensure comprehensive coverage of the topic.

The interview was conducted in two phases. The first phase consisted of exploratory interviews aimed at developing a comprehensive understanding of the research area, Vattenfall's R&D department, and their ways of working. These interviews also helped identify strengths and areas for improvement in project workflows. Participants included Product Owners, Project Members, and experts in AI and data. Simultaneously, observations were conducted within the company to gain further insights. In the second phase, follow-up interviews were conducted at Vattenfall R&D, to gain deeper insights into the employees specific needs and expectations. Finally, in the last phase, the final result and recommendations was presented. Also the analysis of all the information gathered through the project was compiled. In *Figure 5* the timeline of the project is presented.

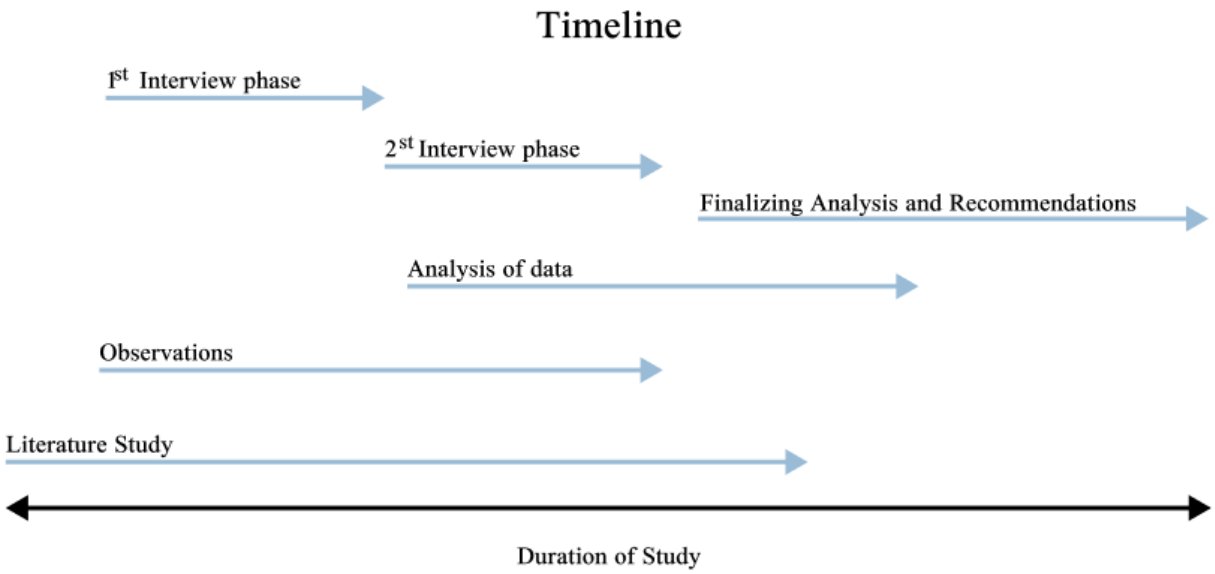


Figure 5: Timeline of the conducted study.

3.4 Literature Study

The literature study was conducted using an iterative approach to develop a theoretical framework, uncover new perspectives, and continuously expand knowledge throughout the project (Snyder, 2019). The literature included articles, reports, case studies, and conference papers. The sources were collected through scientific databases such as Web of Science, Scopus, IEEE, ScienceDirect, and ResearchGate. To find the articles key words as *AI in Agile, Agile project management, Machine learning, AI tools, Project optimization,* were used. As well as manual filtering to find relevant data within the sources, for example publication date.

Initially, the literature study focused on gathering information about agile project management and the role of AI in enhancing it. The primary goal of the first phase was to identify relevant theoretical frameworks to establish a solid foundation for the project. In later phases, the iterative process progressed further, continuously shaping the project's development and refining key areas of interest based on insights from interviews and internal documents.

The foundation of the literature study revolves around three main areas of interest: Agile frameworks, AI-driven project management, and knowledge management in project-based teams. These areas offer valuable insights into how organizations can leverage AI to drive innovation, strengthen agile workflows, enhance knowledge sharing, and improve risk mitigation. The relevance of these themes is underscored by their role in driving continuous learning, optimizing decision-making, and creating an environment that supports innovation and long-term adaptability.

3.5 Data Collection

This study employs four complementary data collection methods: research articles, interviews with employees at Vattenfall, a review of internal documents, and observations conducted throughout the study. This multi-method approach ensures a comprehensive understanding by capturing diverse insights and perspectives.

3.5.1 Interviews

To gain a fundamental understanding of the current situation at Vattenfall, ten 30-minute exploratory interviews were conducted during the first phase of the interviews. A few questions were prepared to point out the strengths and weaknesses of their current project work situation. These questions ensured that the important information was collected. The exploratory interview questions are provided in *Appendix 9.1*.

The participants were selected based on recommendations from Vattenfall's supervisors, who have a strong understanding of the employees and can identify those who would be suitable for the interviews. During the first phase of interviews, individuals who had worked at Vattenfall for a while and were well-versed in the processes and methods were interviewed. These participants were chosen for their deep knowledge and experience within the organization. Additionally, individuals with AI expertise were also interviewed to provide valuable input regarding the use of AI in knowledge management.

For the second-phase interviews, the participants were also recommended by Vattenfall's supervisors, with the intention of ensuring a variation of individuals from different teams to provide a broader perspective in the responses. These 30-60 minute interviews aimed to understand how Vattenfall R&D employees identify relevant projects and stakeholders, what information they need throughout the project lifecycle, and why deviations from the *VF Portal* process occur. The findings contributed to analyzing the agile framework and the role of AI in knowledge management. Additionally, the interviews focused on identifying information needs at various stages of a project, with the goal of supporting the development of a structured documentation process. The second-phase interview questions are provided in *Appendix 9.2*, and an overview of the interviewees and their roles is presented in *Table 3*.

Table 3: Interviewees and their role.

Phase	Respondent	Role	Date
1	A	Product owner	2025-01-22
1	B	Product owner	2025-01-23
1	C	Project member	2025-01-24
1	D	Product owner	2025-01-24
1	E	Brand and AI Strategist	2025-01-28
1	F	Project member + AI/data	2025-01-28
1	G	Head of section	2025-01-29
1	H	Product owner + AI/data	2025-02-03
1	I	Scrum master + AI/data	2025-02-06
1	J	Product owner + AI/data	2025-02-10
2	K	Project member	2025-03-26
2	L	Project member	2025-03-26
2	M	Project member	2025-03-31
2	N	Product owner	2025-04-01
2	O	Product owner	2025-04-03
2	P	Product owner	2025-04-03
2	Q	Product owner	2025-04-07

3.5.2 Internal Documents and Observations

Internal documents from Vattenfall were used to gain a deeper understanding of their way of working. Mainly it was project related documents as old reports or instruction documents for projects, as well as documents about their agile framework.

Observations were conducted through meetings across the R&D department, often their *Weekly Meetings*, this running simultaneously with other parts of the study. This approach provided valuable insights into the actual work processes, allowing for a comparison between observed practices and the responses given during interviews.

This combination of document analysis and observations ensured a more comprehensive and triangulated understanding of the organization's practices, enhancing the validity and reliability of the findings.

3.6 Development

After the initial phase of interviews and literature research, a structured development process was initiated. Insights from literature, internal documents, and interviews were integrated to ensure alignment between theoretical principles and practical applications. An iterative approach, guided by continuous feedback from Vattenfall supervisors and employees, allowed for ongoing refinement to effectively address emerging challenges and opportunities.

To further support this process, a SWOT analysis was conducted to identify key strengths, weaknesses, opportunities, and threats of the final suggested solution (Helms & Nixon, 2010). This provided a structured overview of internal and external factors, helping to refine focus areas and guide the next steps in the research. Additionally, Nonaka's SECI model was applied to gain a deeper understanding of knowledge sharing, while an implementation map (IM) was developed to provide a clear visualization of the current situation and strategic direction (Kang et al., 2024). This was an iterative process that ran concurrently with the second-phase interviews, allowing for continuous refinement based on emerging insights.

The combination of analytical tools and iterative stakeholder engagement was chosen to ensure both strategic depth and practical relevance, allowing the solution to be grounded in real organizational needs while remaining theoretically robust.

3.7 Data Analysis

The qualitative data gathered from two phases of interviews were analyzed to identify key themes and patterns. Prior to the interviews, all participants gave their consent, ensuring ethical standards were met. Each interview was carefully documented to capture all relevant details, which formed the foundation for further analysis. In the initial phase, the notes were thoroughly reviewed to gain a deeper understanding of the respondents' perspectives and their current practices at Vattenfall. The second phase focused on more in depth questions aimed at addressing the research questions. Additionally, data from observations and document reviews were incorporated into the overall data collection to provide a comprehensive analysis. This multi-source approach was selected to enable data triangulation, strengthening the credibility of the findings by cross verifying insights from different types of evidence.

3.8 Quality of Research

To strengthen both the credibility and reliability of this study, a combination of multiple research methods and diverse data sources has been employed. The integration of qualitative interviews, company data analysis, and an in depth literature review provides a comprehensive understanding of the research subject. As suggested by Saunders et al. (2019), utilizing various data sources allows for methodological triangulation. This approach enhances the overall quality of the study by leveraging the strengths of different research methods and mitigating their individual limitations, which leads to more robust findings.

Validity

To strengthen the internal validity of this study, a carefully structured and methodologically rigorous approach was adopted, utilizing qualitative data collection methods (Blomkvist & Hallin, 2015). Data triangulation was applied to validate the information collected from interviews, observations, company documentation, and relevant academic literature. The external validity was ensured by selecting Vattenfall as a case study, as the company represents the typical challenges large energy companies face when integrating innovation and agile methods. This choice allows the study's results to be applicable to other similar organizations within the energy sector (Saunders et al., 2019).

Reliability

Reliability, in the context of this research, refers to the consistency and dependability of the methods and outcomes. To guarantee greater reliability, the study followed a thorough and consistent documentation process for data collection, while qualitative interview data was systematically categorized (Blomkvist & Hallin, 2015). These measures ensure that the research is replicable under similar conditions and would produce comparable results. Furthermore, maintaining a clear and transparent trail of evidence throughout the study allows for the conclusions to be supported and traced back to the original data sources.

However, it is important to note that there are certain limitations to the reliability of the results. Not all questions were identical across all respondents, during phase 1 interviews, which introduces potential inconsistencies in the data. Additionally, the way questions are framed can influence the responses, potentially introducing researcher bias. For example, the tone or phrasing of a question might lead respondents in a particular direction, affecting the objectivity of the answers. To minimize this bias, careful attention was given to structuring the interviews in a way that encouraged open-ended responses and avoided leading questions.

3.9 Ethical Research Considerations

Ethical considerations are crucial in research, especially when engaging with individuals within companies, as demonstrated in this case study at Vattenfall. The study adheres to established ethical standards set by institutional review boards (Saunders et al., 2019). Informed consent is obtained from all participants before the interviews to ensure voluntary participation. During the interviews with company representatives, the purpose of the study was clearly communicated. To safeguard sensitive information, anonymity is preserved, and confidentiality is upheld (Saunders et al., 2019).

4. Results and Analysis

The case study follows an iterative approach, presenting findings in the order they emerged throughout the project.

4.1 Initial Insights and Analysis

Based on the phase one interviews, initial research and findings, an analysis was conducted, resulting in a clustering process to summarize the gathered information.

4.1.1 Strengths of the Current Approach to Agile Project Methodology

The interviewees highlighted several strengths in their way of working, emphasizing their flexibility and the presence of a clear and structured workflow. They also highlighted the benefit of having a high degree of freedom, which enables them to move forward quickly and adapt when needed. In order to validate their direction, they apply hypothesis testing by posing specific questions and evaluating them through binary (yes/no) responses. For example, *"Can X be solved with Y? Yes/No."* The interviewees are also happy with the agile framework, although its application varies between teams. For instance, while some teams prefer frequent meetings to maintain alignment, others prioritize minimizing them to reduce administrative overhead.

4.1.2 Challenges Impacting Project Efficiency and Key Improvement Areas

Despite existing strengths in project workflows, the interviews identified several key challenges affecting efficiency. Addressing these issues could significantly enhance collaboration, knowledge sharing, and project execution.

Time Consuming to Start a New Project

Initiating a project presents a valuable opportunity to set a strong foundation. While it currently involves thorough information gathering, method selection, and verification of data availability, streamlining this phase can lead to more efficient project starts. Interviewees highlighted the potential of leveraging insights from previous, relevant projects early on to accelerate the transition from idea to implementation and enable a smoother and more informed project initiation.

Challenging to Identify Key Stakeholders During the Project

Another significant challenge is identifying and connecting with the right stakeholders. It can be difficult to determine who to contact, and even when the correct person is found, their availability can impact project progress. This underscores the need for a tool that identifies key stakeholders throughout the project lifecycle.

Knowledge Utilization and Collaboration Across Projects

The interviewees emphasized the importance of leveraging existing knowledge and encouraging collaboration across project boundaries, particularly when working on similar initiatives. Today, this is largely managed through close communication between departments to ensure that efforts are aligned and to avoid working on overlapping projects. While this approach is effective to some extent, it also highlights the opportunity to further streamline knowledge sharing. Finding relevant documentation or identifying ongoing and past projects can still be time-consuming, as there is currently no shared system in place. This became especially clear during an observation where an employee expressed the need to connect with someone who had worked on a similar project and to revisit past lessons learned. These insights point to the potential value of a more structured way to access project information and connect with key stakeholders, which could enhance collaboration and reduce reliance on informal networks.

Challenging to Predict Risks and Bottlenecks

There is a valuable opportunity to enhance project efficiency by improving the ability to anticipate risks and potential bottlenecks. In some cases, a limited availability of relevant projects can impact the progress of ongoing work. By leveraging historical data, teams could gain better insights into potential challenges early on, enabling more proactive planning and smoother project execution.

Maintaining a clear focus throughout a project is another area where improvement can drive better outcomes. As teams sometimes approach the *VF Portal* framework differently, their ways of working may vary - something that presents an opportunity to further align methods and strengthen collaboration. With clearer structures and shared practices, teams can stay more consistently on track and focus their efforts on the project's core goals.

Receiving Assistance to Ensure Necessary Documents Are Completed Correctly

Documentation is sometimes deprioritized in agile workflows due to the risk of excessive administrative work. However, interviewees emphasized that proper documentation remains essential, as past experiences can be invaluable for future projects. They mention that easier access to historical project data would provide significant benefits for learning and efficiency. It would also increase motivation for documentation, as the information would be reused.

There is a clear opportunity to improve document management by introducing more structure and accessibility. A more organized system would make it easier for teams to quickly locate the files they need, supporting smoother workflows and more efficient collaboration. The interviews revealed gaps in documentation practices, leading to inconsistency across teams. One key issue is the lack of a system to flag errors in documentation, despite the existence of guidelines. For example, when it comes to archiving, teams occasionally miss the process, which can result in documents remaining in project folders that are not accessible to everyone. This becomes especially problematic for projects that need to be archived before being shared among team members due to confidentiality requirements. These challenges highlight the need for a system that offers guidance and ensures proper documentation, facilitating efficient information reuse.

Security classifications add another layer of complexity, one of the key challenges in implementing AI at Vattenfall is ensuring the security and confidentiality of information. All their information is classified from public to strictly confidential. This sometimes makes it hard to share information, even between employees at the R&D department. So one challenge will be to implement an AI without sharing any confidential information with the ones that can't have access to that data. All documents at Vattenfall get the classification, so it is clear what can be shared, but in some cases the project reports need to be rewritten and some information taken away before archiving them. Furthermore, it is crucial to comply with the General Data Protection Regulation (GDPR) in the European Union. In short, this means that no unnecessary personal information should be collected or stored.

4.1.3 Enhancing Project Efficiency and Documentation through AI and Process Optimization

The interviewees expressed a strong desire to continuously improve project efficiency by streamlining workflows and reducing manual tasks. They emphasized the importance of simplifying stakeholder interactions, minimizing administrative burdens, and ensuring easier access to relevant data.

A theme in the interviews was the potential role of AI in project management. Employees highlighted how AI could support various tasks, such as structuring and retrieving information, preparing for meetings, transcribing discussions, and summarizing key takeaways (which is current Microsoft Copilot functions). AI could also help capture critical insights that are often lost, such as valuable discussions in email threads and informal meetings that do not make it into final reports. For instance, AI-driven solutions could enhance project initiation by compiling relevant data at an early stage, making the transition from idea to execution more efficient.

However, a key challenge must be addressed before AI can be effectively leveraged in documentation and project management: the current lack of sufficient and structured documentation. To enable AI tools to provide meaningful support, clear guidelines and consistent practices are essential. The interviewed AI experts emphasized the importance of standardized report templates and validation mechanisms to ensure high-quality documentation. Without well maintained documentation, the potential of AI to add real value remains limited.

Building on this, another important consideration is the need to distinguish between inefficiencies stemming from agile workflows and those unrelated to agile methods. Clarifying this distinction can help refine project management practices and ensure that AI-driven solutions are addressing the right problems.

4.2 Knowledge Sharing and Documentation in Practice

A clear understanding of the tool's purpose and the type of information users need was essential for its development. To gain these insights, a second phase of interviews was conducted to explore what users typically seek in past projects and how they identify key stakeholders.

The interviewees were divided into two target groups: employees with the role of Project Member (PM) and those with the role of Product Owner (PO). These groups were chosen to gain a deeper understanding of how they perceive AI, knowledge sharing, and documentation based on their respective roles and responsibilities within projects. Generally, POs hold greater responsibility and play a key role in ensuring that documentation and related processes are carried out correctly in their projects.

4.2.1 Current use of AI

The Project Members reported using AI regularly, mainly as a tool for quickly resolving coding related issues. This includes instances where they need to program in an unfamiliar language or have forgotten a specific command. At the start of new projects, AI also supports them in efficiently reviewing articles and gaining a high level understanding of the current state of the field. Overall, they described AI as a valuable time saver.

The Product Owners, meanwhile, use AI more extensively. Beyond coding support, they engage with AI as a discussion partner, quickly obtaining technical answers and references. For example, while awaiting input from subject experts, they use AI to delve deeper into technical topics. This has proven especially helpful when working with simulations and aiming to better understand technical systems, areas that may lie outside their core expertise but are important to grasp.

A notable advantage they highlighted was avoiding the advertisements and distractions often associated with traditional search engines. They also emphasized AI's usefulness for summarizing reports and crafting short descriptive texts.

One of the more senior Product Owners mentioned that they had not used AI extensively in the past, primarily due to a lack of familiarity. However, after recently completing a course on Copilot, they expressed excitement about exploring AI further and integrating it more actively into their work.

4.2.2 Successful Knowledge Sharing in the Organization

The interviewees shared several examples of successful knowledge sharing in their organization. One person described how they discovered a previous project similar to their own, which allowed them to reuse parts of the code and significantly reduce development time. Another noted the value of meetings in identifying the right colleagues to contact for specific questions. A third shared how they came across an old report that turned out to be directly applicable to their current project.

A particularly successful example came from a demo day, where extra effort was put into presenting a project in a pedagogical and accessible way. This approach made it easier for colleagues from various backgrounds to follow along, which led to more questions than usual and, in turn, richer feedback. This improved both collaboration and learning within the team.

All interviewees emphasized the importance of archiving past projects. Having access to previous work was seen as a valuable way to build on existing knowledge and avoid reinventing the wheel.

When asked what kind of documentation is most useful for ongoing and future projects, the interviewees pointed to several key elements. At the start of a project, they often search for similar past projects to understand which methods, code, and approaches were used. Insights into what worked well, and what did not, were considered crucial to avoid repeating mistakes and to apply best practices moving forward.

During a project, the interviewees emphasized the importance of capturing “rough drafts” of decisions, notes explaining the reasoning behind coding choices and design directions. This type of documentation helps others (and their future selves) understand the context behind key decisions.

In terms of knowledge sharing practices, the Project Members mentioned that they primarily exchange information through meetings. Important information is also expected to be documented and stored in SharePoint, ensuring easy access for the entire project team. Additionally, completed projects are archived in *Kollektiva Minnet* for long-term reference.

Meetings and regular check-ins were seen as critical to effective knowledge sharing. Interviewees also used Microsoft Teams to share relevant research articles and updates. However, much of the knowledge exchange still happens informally. To foster collaboration and reduce redundant work, some teams hold joint monthly meetings with other teams to stay updated and learn from each other. In one case, a member from a related team occasionally joined another team's meetings to ensure alignment and prevent overlapping efforts.

4.2.3 Future Vision of AI to Enhance Knowledge Sharing

The Project Members recognized the significant potential of AI to enhance knowledge sharing. They envisioned AI simplifying the process of searching through past projects, making it faster and more efficient to find relevant information. This would reduce the time spent manually navigating through previous work and improve access to valuable insights, facilitating smoother project execution.

Similarly, the Product Owners underscored the value of revisiting past projects. They suggested that specifications from previous assignments could be revised and reused, rather than starting from scratch, thus optimizing time and resources. Old reports could also serve as useful tools for onboarding new team members, offering them valuable context and guidance. However, the POs pointed out that when a new project diverges significantly from previous ones, a deeper understanding of the end user becomes even more critical to ensure its success.

Throughout the project lifecycle, the Product Owners stressed the importance of documenting decisions and the rationale behind them. Without this documentation, there is a risk of re-testing elements or losing track of why certain decisions were made. They also emphasized the need to document "showstoppers", critical blockers that could impede project progress. By capturing these key moments, future teams could avoid repeating mistakes and better anticipate challenges.

For documentation to be truly valuable in future projects, it needs to be well organized and contain actionable insights. For example, if a team realizes that more frequent meetings with a supplier could have prevented delays, this should be clearly documented, along with the consequences of not doing so. Such details would help future teams apply lessons learned in a meaningful way, improving overall project efficiency.

Looking ahead, interviewees expressed a strong desire to improve the documentation of project work, particularly the steps unique to Vattenfall. They emphasized the importance of clearly outlining processes like requesting access to different types of data, which would greatly benefit new employees who are unfamiliar with internal procedures. While these steps may seem obvious to long-time employees, they are not always intuitive to others. There was also a call for updated frameworks that describe Vattenfall's operational practices, including common challenges, unexpected obstacles, and the solutions used to overcome them. Capturing what has worked well and identifying best practices is essential for ensuring continued success in future projects.

Many interviewees expressed a desire for a unified approach to working that could be consistently followed across all R&D teams. They also shared a positive outlook on AI's potential to support knowledge sharing. AI could streamline the process of searching for internal documents and information, eliminating the need to manually sift through multiple files. Overall, the interviewees believed that AI could play a crucial role in retrieving old information and compiling relevant insights quickly, enhancing the overall efficiency of the knowledge-sharing process and facilitating smoother collaboration across teams.

4.2.4 Identified Challenges Related to Knowledge Management in Current Workflows

The interviewees identified several recurring challenges related to knowledge management, collaboration, and the use of existing tools and processes within the organization.

Difficulties in Identifying and Accessing the Right Stakeholders and Documentation

A common issue reported by the interviewees was the challenge of locating and contacting the appropriate stakeholders during projects. These individuals often possess specialized technical or organizational knowledge that is critical to project success. However, it is frequently unclear who holds the relevant expertise, forcing project members to rely heavily on the Portfolio Manager for guidance. This reliance creates a knowledge bottleneck, where access to critical information becomes overly dependent on a single individual, increasing organizational vulnerability.

Moreover, challenges were also highlighted regarding the retrieval and reuse of documentation, particularly when revisiting older projects where original team members are no longer available. In such cases, a significant amount of contextual knowledge is lost, making valuable insights difficult to retrieve. While experienced employees may become better at navigating documentation systems over time, the process remains complex and cumbersome. The Product Owners noted that they occasionally recommend past projects as references, but it is the Portfolio Manager who holds a comprehensive overview of previous projects and expert competencies. The lack of transparency regarding who holds which knowledge adds an additional layer of difficulty in accessing necessary information, further complicating knowledge sharing and reuse within the organization.

Risks of Over Reliance on AI Tools

While AI tools are appreciated for their ability to provide quick answers, one interviewee raised a concern about their potential impact on long term learning. When users consistently turn to AI for solutions without fully engaging in the problem-solving process, they may not internalize the knowledge. For example, frequently relying on AI to recall specific commands can lead to repeated lookups, reducing efficiency over time and undermining the intended time saving benefits.

Inconsistent Meeting Documentation and Feedback from Stakeholders

Another identified gap was the inconsistent or missing documentation of meeting notes, which could otherwise serve as valuable references throughout the project lifecycle. Additionally, there is a notable lack of documentation on how end users, whether internal or external stakeholders, experienced the project outcomes. This represents a missed opportunity for feedback that could inform future project improvements.

Low Utilization and Perceived Rigidity of VF Portal

The interviewed Project Members reported limited familiarity with the internal project documentation tool, *VF Portal*. Several admitted to never having used it themselves. Instead, knowledge tends to be shared through informal channels such as presentations, team chats, or as old reports. The Product Owners attributed the low adoption of *VF Portal* to its perceived rigidity and lack of flexibility. They described the system as overly structured and unsuitable for the fast paced and iterative nature of their work. As a result, teams often develop their own processes and formats, although more practical, lead to inconsistencies across the organization. A further reason for the tool's underutilization is the absence of adequate onboarding or training. In many cases, project teams are under time constraints, and the documentation requirements in *VF Portal* are seen as too complex and time consuming.

Organizational Complexity and Structural Barriers

Beyond tool specific concerns, broader structural challenges were also brought up. For instance, initiating a new project at Vattenfall was described as a lengthy and highly regulated process, making simplification and flexibility in administration a high priority for teams. In some cases, when projects are conducted on behalf of another department, documents and data must be managed in separate systems, adding further complexity to collaboration and documentation workflows.

There are also some challenges according to agile methodology. Several interviewees discussed the difficulty of applying agile working methods within the current R&D context. Team members are often spread across multiple projects and departments, making it challenging to build cohesive, well-functioning teams. The existing organizational structure and work methods were sometimes seen as misaligned with agile principles, hindering efforts to increase responsiveness and adaptability. They mentioned using "Vattenfall Agile," an unofficial methodology that incorporates an agile approach with a unique personal twist.

Below in *Table 4* is a summarization of the interviews presented.

Table 4: Summarization of Interviews.

Strengths in work processes	Flexibility, clear structure, and high freedom to adapt quickly. Hypothesis testing is used to validate direction through yes/no questions. Agile methods are applied differently depending on the team; some prefer frequent meetings for coordination, while others minimize them to reduce administrative workload.
Challenges and areas for improvement	• Project Kick-off: Time-consuming to initiate projects; there's a need to leverage previous project data for smoother starts. • Identifying Key Stakeholders: Difficult to find the right stakeholders during the project. • Knowledge Sharing: Difficulty in locating old documents and tracking ongoing projects that could collaborate. • Risks and Bottlenecks: It's hard to predict risks; sometimes projects are affected by resource shortages. • Documentation: Better systems are needed for documentation and structure to facilitate future learning and reduce errors.
Security and confidentiality	Classification of information and GDPR require careful assurance of confidentiality when using AI and sharing information.
Vision for future AI use	AI can improve knowledge sharing by quickly finding relevant information from past projects and enhance the quality and accessibility of documentation. However, more unified working methods and better documentation practices are needed for AI to be effective.
AI usage	• Project Members use AI for coding support and reviewing articles. • Product Owners use AI extensively for technical answers, summaries, and reducing distractions from traditional search engines.
Knowledge sharing and documentation	• Previous Projects: Utilize old projects and documents to avoid "reinventing the wheel." • Decision Documentation: Capture decisions and their rationale to aid future projects. • Information Exchange: Important meetings and SharePoint are used to share information, but much still happens informally.

5. Suggested approach to implement AI in agile project management

This chapter presents a suggested approach for integrating AI into agile project management to enhance knowledge sharing and drive innovation.

5.1 Implementation Mapping

Implementation Mapping (IM) is utilized to ensure the successful application of interventions in practice. In this case, it was developed as a strategic approach to guide the deployment of AI for enhancing knowledge sharing and driving innovation.

This chapter presents the proposed implementation roadmap, designed with Vattenfall's strict confidentiality requirements in mind. Due to these constraints, data could not be freely shared or uploaded. To determine what could be implemented at each stage, an implementation mapping was conducted, taking into account the available data. Given that historical data was inaccessible and stored in various formats across different locations, the most practical approach was to assume the tool would be developed incrementally. This strategy proved more convenient, as retrieving and analyzing past projects would have been too time consuming and expensive.

This roadmap will require ongoing review and evaluation to ensure it effectively meets its intended objectives. While the potential applications of AI are extensive, it is equally important to consider the possible adverse implications of its implementation, particularly in the absence of confidentiality constraints. In *Figure 6*, the correlation between implementation time and the amount of stored data is illustrated, along with how this relationship aligns with the different phases of the implementation. Where phase A requires the least amount of stored data and has the shortest implementation time

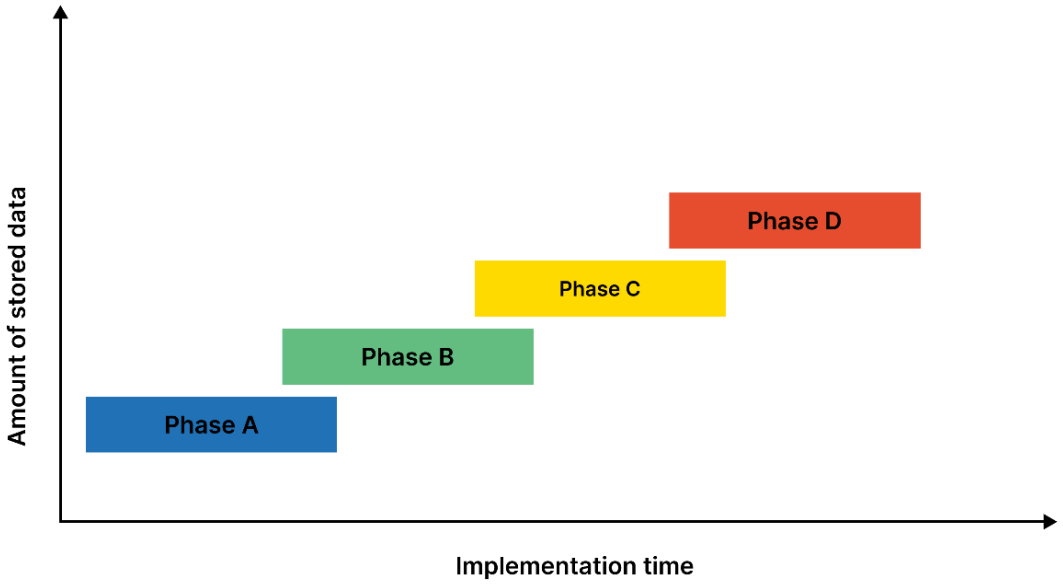


Figure 6: Correlation between stored data and implementation time.

Phase A

Phase A is the functions that can be immediately implemented and that are not dependent on historical project data. During this phase the functions are:

- Suggesting ongoing projects that closely align with one's own, making it easier to find relevant initiatives and potential collaboration opportunities.
- Enabling searches based on specific competencies, providing recommendations on which employees to consult based on their expertise and involvement in ongoing projects.

Phase B

In phase B, additional functions can be introduced as the AI system gains more historical data:

- Enhancing searchability in historical project data to easily access and utilize previously documented knowledge.
- Suggesting past projects that closely align with one's own, making it easier to find relevant insights and learn from previous experiences.
- Enabling searches based on specific competencies, providing recommendations on which employees to consult based on their expertise and involvement in past projects.

Phase C

In phase C, with further additional data, the AI can offer functions as:

- AI analyzes the ongoing project's description to identify potential risks and trends based on past projects, providing recommendations on key factors to consider.
- AI-driven dashboards aggregating relevant insights from multiple projects, providing personalized recommendations - such as newly documented lessons learned that are relevant to the user's project and should be reviewed.
- AI tool for ensuring standards by verifying that mandatory project documents meet required quality criteria. This ensures they can be effectively utilized for future AI-generated insights and prevents projects from being archived unless all necessary documentation is complete.
- AI-powered knowledge hubs that automatically suggest team collaborations and foster spontaneous knowledge-sharing interactions, improving real-time communication and leveraging team expertise.

Phase D

In the final phase, the AI will heavily rely on historical data, making this the last step to be implemented once sufficient data has been accumulated. This phase will include:

- AI-powered risk prediction and error identification by analyzing the uploaded project description. The AI estimates potential risks and suggests optimizations based on insights from past projects, helping teams proactively address challenges.
- AI-driven identification of best practices based on historical project performance. The AI analyzes past projects and then suggests effective strategies and actions, helping teams proactively manage and mitigate challenges by applying proven methods and approaches.

5.2 Required Documentation for Implementation

To ensure a gradual implementation, Vattenfall needed to secure the necessary documentation for each phase. The next step was therefore to closely examine phases A and B to determine the documentation required by the R&D department for their implementation. The process began with a thorough analysis of phase A, assessing how to proceed to ensure it could be implemented immediately and identifying the essential information documented to move forward with phase B.

Phases A and B were the most critical to understand in depth. By analyzing them, along with insights from interviews, conclusions were drawn regarding the necessary documentation to leverage AI in delivering the following outcomes. Phases C and D were excluded from the current analysis, as they were considered dependent on the seamless execution of phases A and B, and were to be implemented at a later stage.

The required documentation for both ongoing and in the future past projects will be:

- Project description
- Lessons Learned, Risks, Results and other documents connected to the project
- Employees involvement in projects
- Employees competencies and areas of expertise

It is difficult to specify exactly what needs to be documented for each of these areas as this needs to be tested, but the following provides a general direction for the necessary documentation to achieve the functions in Phase A and B. By testing this solution in the future, it will become clearer what information is required to effectively connect these elements. This is because the information must be adapted and adjusted to the algorithms.

For example, using sustainability as a keyword would be ineffective, as it would match almost every R&D project. It is only through working with the AI model and testing its outputs that one can determine which keywords and data points will lead to successful recommendations. These test generations can be conducted using archived projects while gradually onboarding new projects into the system.

From the interviews, several aspects were identified as important for employees to have documented. These could be valuable to consider during the development of this AI tool. These things are presented in *Table 5*.

Table 5: Important aspects to document.

During the project	After the project	Vattenfall-specific aspects
• Methods used • What worked well and what didn't • Decision-making logic and its consequences • Showstoppers – critical obstacles that affected the project's progress	• Results and key learnings • Feedback from recipients/end users • What worked well in the process	• Internally adapted processes • Description of common challenges and how they are handled

Below in *Table 6* is an overview of the required documentation to fulfill the functions mentioned in phase A and B.

Table 6: Functions and the required documentation to fulfill the functions.

Function	Required Documentation
Phase A	
Suggest ongoing projects that closely align with one's own	Current project descriptions, objectives, sub-areas, employees' involvement in ongoing projects
Enable searches on employees based on specific competencies	Employees' competencies, areas of expertise, current involvement in ongoing projects
Phase B	
Enhance searchability in historical project data	Historical project descriptions, results, lessons learned, success factors, risks, and failures
Suggest past projects that closely align with one's own	Historical project, descriptions, results, lessons learned, success factors, related projects
Enable searches on employees based on specific competencies	Employees' competencies, areas of expertise, and involvement in past projects

5.3 Conceptual Solution Design

The idea is to input project information, which will generate insights mentioned in phase A and B. Initially, when the AI system is implemented, it will primarily provide insights into similar ongoing projects, along with the employees involved and their areas of expertise. This will facilitate knowledge sharing and idea exchange while also offering a more efficient way to connect with colleagues based on their expertise and project involvement.

Over time, as more data is continuously input, additional features from Phase B will become available. This will enable linking projects not only to similar ongoing initiatives but also to past projects, allowing for broader knowledge access, reduced redundant work, and enhanced knowledge sharing. Additionally, users will be able to identify and connect with individuals who have worked on similar past projects. The system will also support advanced search functionality within archived projects and project descriptions, further enriching the available knowledge base.

This solution assumes that no historical project data is initially provided, except for training the AI model. If past project data were to be included from the start, Phase B could be implemented simultaneously with Phase A, significantly accelerating the system's capabilities and value. Ultimately, this will create a comprehensive and easily accessible knowledge hub that streamlines information sharing, improves accessibility, and strengthens collaboration by connecting employees more effectively.

Based on above, the conceptual AI bot, utilizing the currently available input data as illustrated in *Figure 7*, demonstrates how such a solution could function in practice by generating the corresponding output. This serves as a practical example of how the assistant can streamline workflows, provided that the right conditions for success are met.

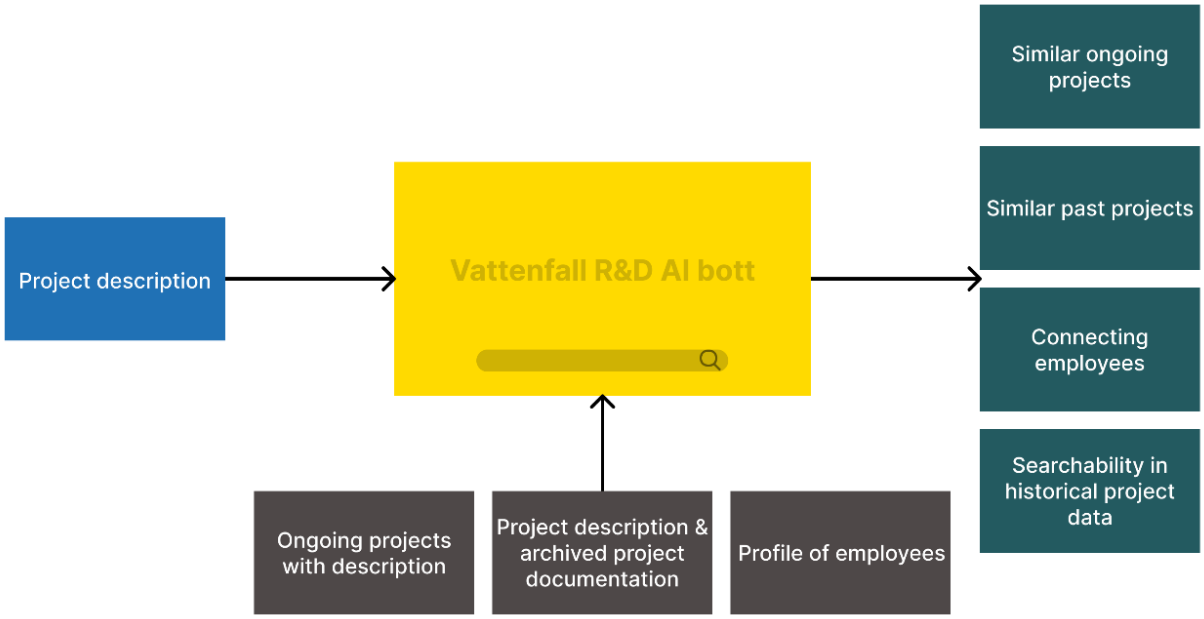


Figure 7: Conceptual AI bot.

5.4 Visual Solution

This visual solution illustrates an example of how an AI driven platform with these capabilities might function. It showcases how AI can support the documentation process by suggesting answers, providing clear guidance for each section and asking follow up questions to ensure completeness and compliance with documentation standards.

Additionally, the design focuses on providing seamless access to essential information before and during a project, ensuring that employees receive the right insights with minimal effort, avoiding unnecessary clicks or navigation. *Figure 8* presents the *Home* page. On the left side, employees can select the project they are working on and upload relevant documentation to receive more tailored AI recommendations. In the center of the homepage, they can interact with the AI bot to ask questions. If project information has been uploaded, there's no need to explain the context when asking specific questions related to the project. On the right side, employees can explore similar projects, both ongoing and completed, for additional insights.

Over time, the homepage could offer increasingly personalized recommendations. For instance, in later project phases such as C and D, the system could highlight potential risks to be aware of based on past experiences from similar projects. Under the page *About* there will be clear instructions for the AI tool, with supporting videos and guidelines for usage.

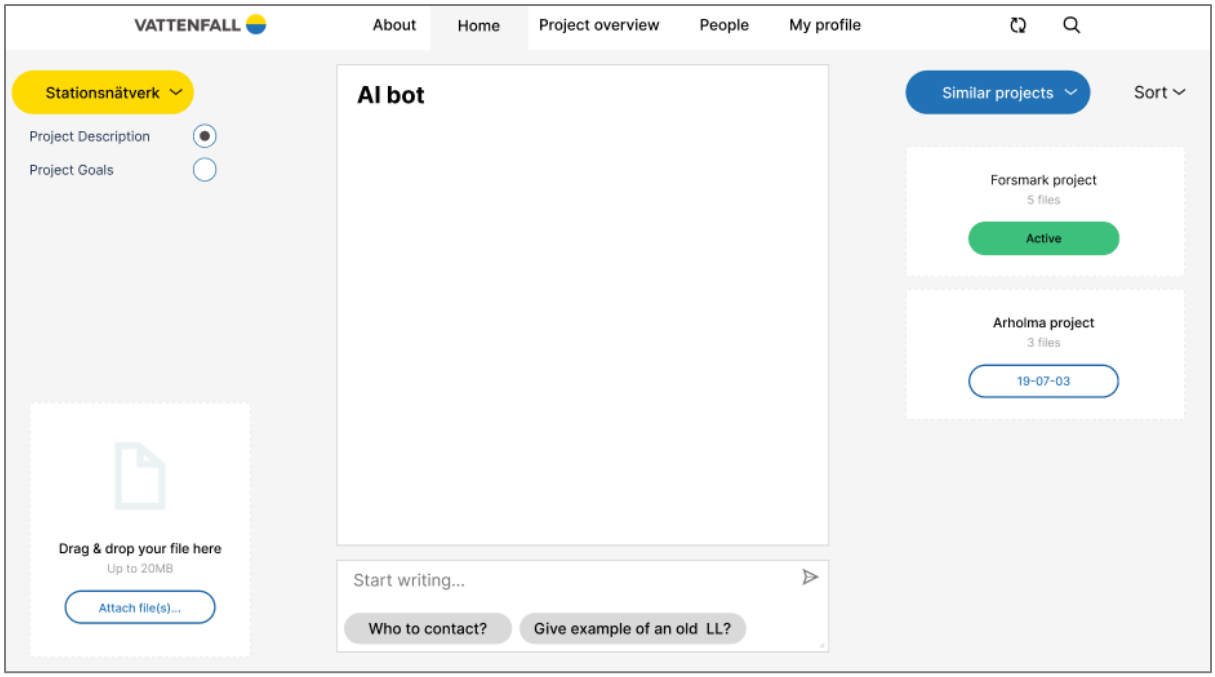


Figure 8: Home page.

The next page *Project overview*, in *Figure 9*, displays the projects you are involved in along with their current status. This layout can be adapted depending on the user. For example, a portfolio manager could get a comprehensive view of all projects within their portfolio. In the long run, it could also offer statistics on how many projects have been carried out in different areas. This would serve as a smooth and efficient way to maintain an overall perspective on project activities.

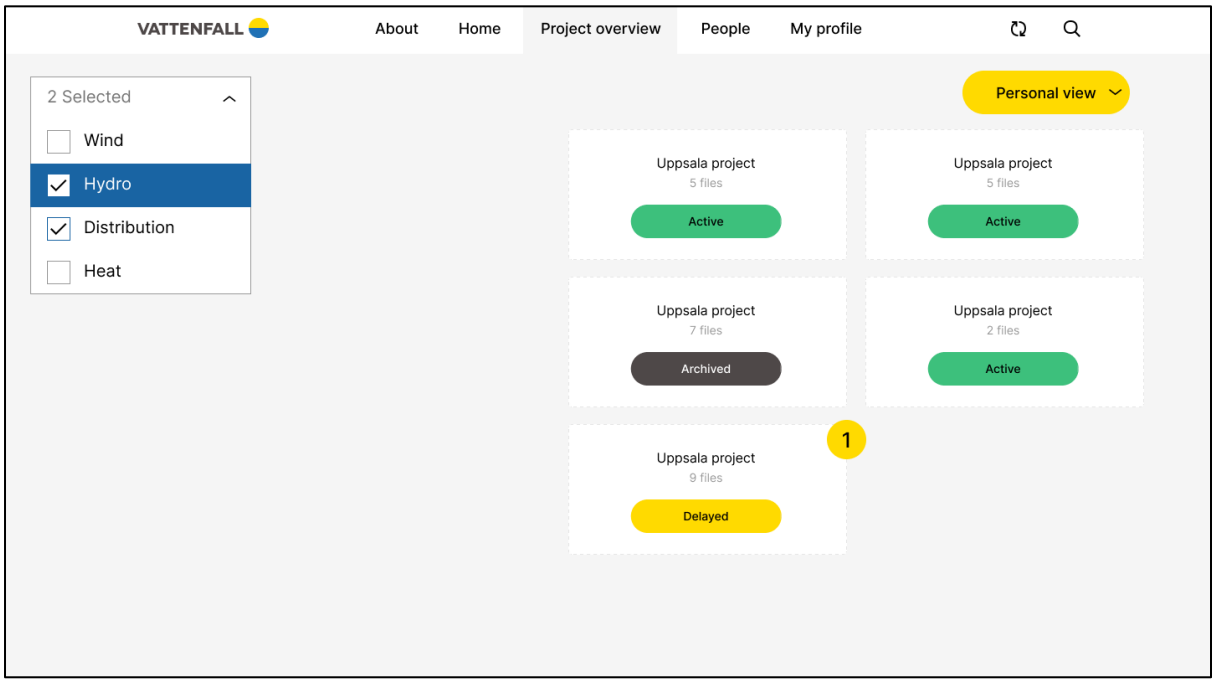


Figure 9: Project overview.

When clicking into a specific project, users can view all uploaded documents as well as messages outlining what needs to be completed before the project can be archived, this Illustrated in *Figure 10*. In this view, users can also chat with a bot that provides guidance on the steps required to finalize the archiving process. Additionally, the bot provides guidance on how documents should be structured and formatted to meet the necessary requirements. This helps minimize the risk of projects not being archived properly and increases the likelihood that valuable knowledge can be reused.

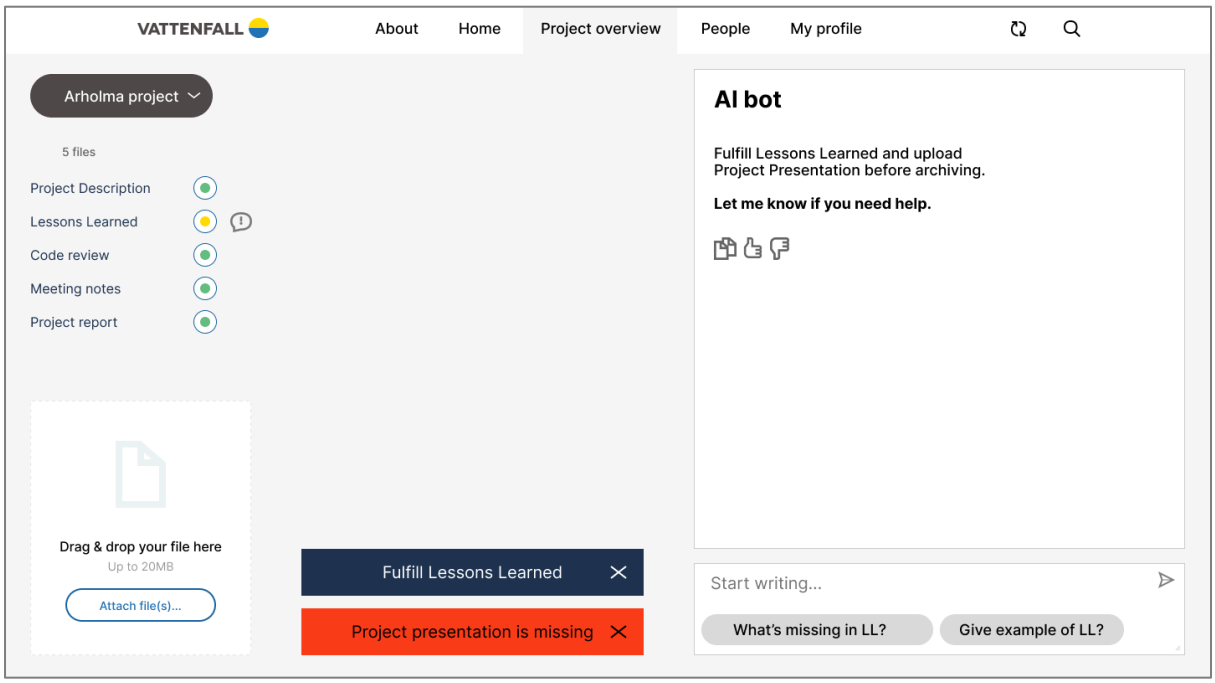


Figure 10: Specific project.

Under the *people* section, in *Figure 11*, employees can search for relevant colleagues by filtering based on competencies, project experience, and department within Vattenfall. The system then provides tailored recommendations of individuals they can connect with, making it easier to find the right expertise and foster collaboration.

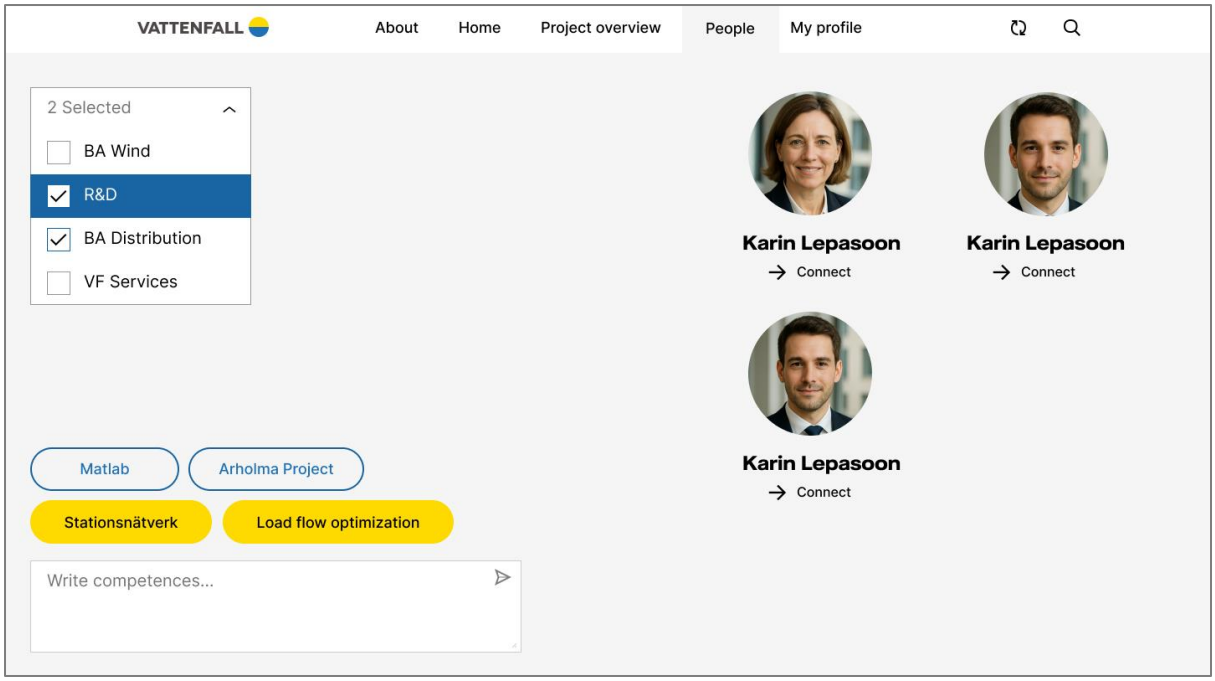


Figure 11:People.

Under *My Profile*, each employee can edit their listed competencies and update which projects they've been involved in. They can keep their profile up to date with the latest information and connect it to Teams, making it easy for others to reach out and get in touch, this in *Figure 12*.

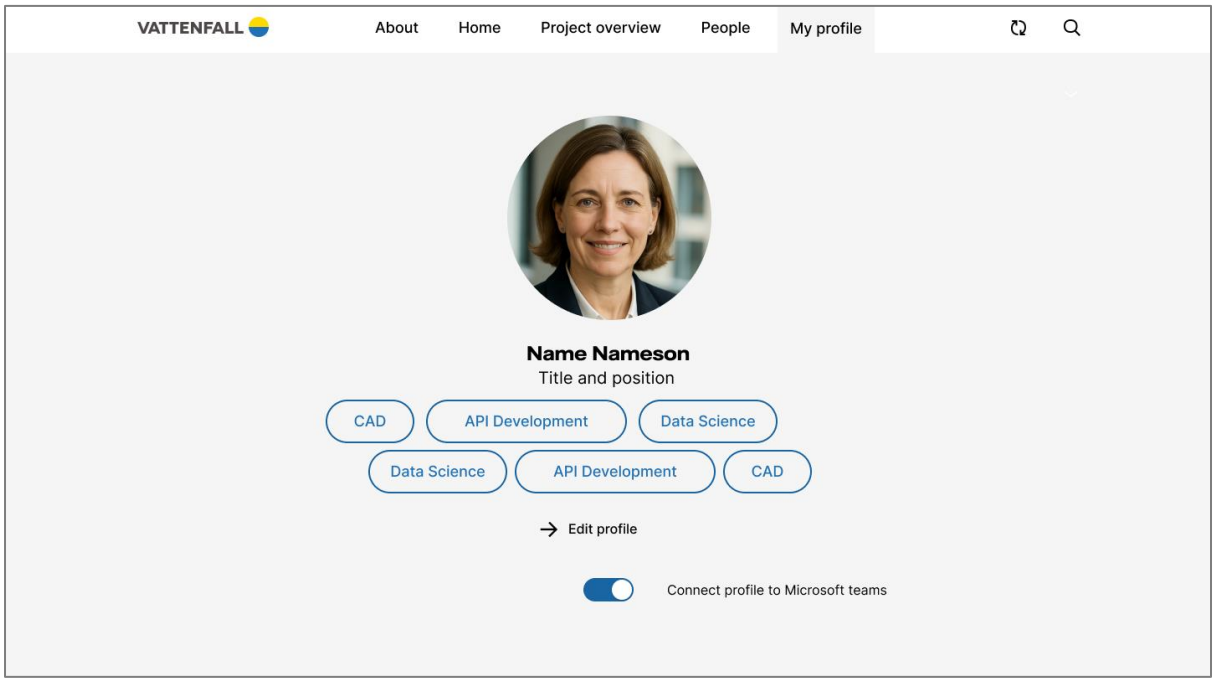


Figure 12: My profile.

Other features identified as highly valuable based on interviews include AI-assisted support in ensuring document completeness. Since certain mandatory documents must be filled out correctly throughout the project lifecycle, an AI assistant that asks follow-up questions to ensure completeness can be extremely helpful. This would guarantee that all required documents are in place while also laying the foundation for the eventual implementation of Phases C and D, where the AI will rely more heavily on comprehensive data to draw conclusions and provide recommendations. Additionally, this would reduce the workload for individuals who currently review and assess these documents, as submissions would meet a higher standard and maintain greater consistency, regardless of the author, thanks to AI-driven support.

5.5 SWOT Analysis

Based on the visual and conceptual solution, as well as the implementation mapping, a SWOT analysis was conducted, focusing on finding the strengths, weaknesses, opportunities and threats, with implementing this suggested AI tool. This summarized in *Figure 13*.

Strengths: One of the key strengths of this AI-driven system is its phased implementation, allowing for early functionality while continuously improving over time. By connecting employees based on expertise and involvement in ongoing projects, collaboration and knowledge sharing are significantly enhanced. Additionally, the AI provides searchability in historical project data, making it easier to access relevant insights and identify similar past projects. As the system matures, it will contribute to better risk management and optimization, identifying potential issues, and suggesting best practices to enhance project outcomes.

Weaknesses: A major weakness of the system is its heavy reliance on historical data, meaning that full functionality will only be achieved after a significant amount of data has been collected. This creates a long setup period before the AI can deliver its most valuable insights. Furthermore, integration challenges with existing systems may arise, requiring technical adjustments and resources. User acceptance is another potential issue, as employees might be hesitant to trust AI driven recommendations or change their workflows. Lastly, the quality of input data is crucial, if the data is inconsistent or incomplete, the AI's insights could be misleading or inaccurate.

Opportunities: The AI system presents several exciting opportunities. By automating analysis, risk detection, and code review, it can significantly boost productivity and reduce manual workload, allowing employees to focus on higher value tasks. Additionally, the AI driven learning platform can provide personalized training recommendations, helping employees develop new skills and enhancing overall organizational competence. The system is also highly scalable, meaning it can be expanded to cover more business areas and projects in the future. Organizations that successfully integrate all these steps of AI into project management and product development may also gain a competitive advantage, as in the long run, data driven decision making and efficiency improvements can lead to better project outcomes and increased market positioning**.**

Threats: Despite its potential, the AI system also faces certain risks. Data security and privacy concerns are critical, as handling sensitive project data requires robust protection measures to prevent leaks or unauthorized access. Regulatory challenges, such as compliance with GDPR and other data protection laws, could also limit how AI can process and analyze data. Additionally, there is a risk that organizations may become over reliant on AI insights, which could lead to poor decision making if the system provides inaccurate or biased recommendations. Finally, high development and maintenance costs could pose a financial challenge, requiring long term investment to ensure continued functionality and improvements.



Figure 13: Summarization of SWOT analysis.

The results of the SWOT analysis underscore the importance of considering all aspects of AI implementation to ensure long term sustainability. While the AI tool offers significant strengths and opportunities, it also presents certain weaknesses. All the external threats must be carefully addressed, as overlooking any of these factors could hinder the successful adoption and sustainability of the AI solution.

6. Discussion

This chapter discusses the findings and contributions in relation to the research question and objectives.

This study aims to address the following research questions:
RQ1: What applications of AI can facilitate knowledge retention and accessibility in an agile environment?
RQ2: How can AI be integrated to support agile project management to strengthen knowledge sharing as a foundation for innovation?

To answer these questions, the study provides both an implementation roadmap for integrating AI tools (specifically LLMs) into Vattenfall's agile project workflows and a visual solution that illustrates how such tools can support knowledge retention and sharing in practice. In addition, the findings are analyzed in relation to existing literature. While the primary goal has been to support Vattenfall with practical recommendations. The study also contributes to the academic discourse by identifying insights that are not yet sufficiently addressed in current research.

This study makes several contributions to the understanding of how AI can be integrated within agile frameworks, an area that has received limited attention in prior research. First, it addresses the paradox between AI's need for structured data and agile's preference for minimal documentation, highlighting the conflict between AI's reliance on clear, organized input and agile's emphasis on flexibility and minimal documentation. Second, the study identifies a significant gap in the current literature, the lack of AI implementation in agile project management. Despite the growing potential of AI, its role in enhancing agile methodologies, particularly in fostering innovation and facilitating knowledge sharing, remains largely unexplored. Lastly, the study contributes to the emerging literature on AI and knowledge sharing by examining the potential of integrating AI with the SECI model. A well-known framework for knowledge management, to enhance knowledge sharing and collaboration across teams. By optimizing the stages of the SECI model, AI can facilitate continuous learning and more effective decision-making within agile environments. Together, these contributions offer valuable insights and directions for future research, enriching the understanding of how AI can be effectively leveraged to support agile practices.

6.1 The Paradox of AI and Agile: Navigating the Documentation Conflict

This study reveals that the integration of AI into agile environments presents a paradox. AI systems rely on structured, high-quality documentation to function effectively and ensure fair, explainable outcomes. Agile methodologies on the other hand, traditionally deprioritize extensive documentation in favor of adaptability, speed, and continuous delivery (Zaitsev et al., 2018; Azam et al., 2024).

This conflict becomes particularly evident when agile teams, focused on rapid iteration and flexibility, prioritize working solutions over detailed documentation. This leaving AI systems with inconsistent or incomplete inputs and results in AI being either underutilized or less effective, as it relies heavily on structured data to generate accurate insights.

This tension creates challenges for organizations, where they need to find a way to maintain the light, iterative nature of agile development while creating the documentation required for AI to deliver reliable support. A potential solution lies in leveraging AI not just as a passive recipient of documentation but as an active enabler. By integrating AI into agile workflows, documentation can be generated more easily and smoothly, reducing resistance from development teams.

This approach is reflected in the implementation roadmap, where AI functionalities progressively evolve from recommending relevant ongoing projects and competencies (Phase A) to enabling access to and analysis of historical data (Phases B–D). By embedding these capabilities into existing workflows, the system aims to reduce the perceived burden of documentation, ensure the quality and completeness of project records, and enhance knowledge reuse across projects. As the analysis shows, by strategically integrating AI to support documentation efforts, it becomes possible to meet the informational demands of AI systems without compromising agile principles. For instance, AI can assist in capturing relevant data and project insights in real time, transforming routine processes such as meeting notes, sprint reviews, or summaries into structured, accessible knowledge bases (Fairfax, 2024).

Furthermore, by integrating the AI solution with tools such as Microsoft Copilot, it becomes possible to automate administrative tasks, thereby streamlining knowledge capture and reducing the manual workload. By seamlessly embedding these AI capabilities into agile workflows, the system supports speed and flexibility while simultaneously enhancing knowledge sharing, and long-term learning across projects.

A key consideration emerging from the investigation is whether AI should be applied selectively, for example, in project evaluation phases where high-quality documentation is essential for drawing reliable conclusions, or whether it should be embedded throughout the entire project lifecycle. The latter approach offers the advantage of maintaining agility while ensuring that critical insights are continuously captured. If doing so, documentation evolves from being a static requirement to becoming a dynamic and integrated part of the development process, supporting iterative progress without undermining transparency or organizational learning.

6.2 Bridging the Gap: The Underutilization of AI in Agile Project Management

In today's rapidly advancing technological landscape, effective project management is crucial for driving and sustaining innovation (Henry Oziegbe Iriogbe et al., 2024). However, the increasing burden of manual tasks often leads to deprioritization of documentation by project members, resulting in inconsistencies and gaps in crucial project information.

Integrating AI into project workflows provides a powerful solution, enabling organizations to automate routine tasks, enhance efficiency, reduce costs, and improve overall quality. Without the adoption of AI-driven solutions, companies risk missing valuable opportunities for process optimization and resource savings. Leveraging AI to streamline project management allows for more effective alignment with strategic objectives, ensuring both immediate gains and long-term success (Fairfax, 2024).

The growing complexity of innovation projects and the increasing need for efficiency highlight the importance of exploring how new technologies can improve agile project management practices. This study contributes to existing research by addressing the challenges associated with the agile approach to project management and explores how AI can offer solutions to these issues. It builds on the increasing need for effective knowledge sharing in complex innovation projects and emphasizes the potential of AI to enhance efficiency and collaboration.

While previous studies have focused on automation at a general level, this study delves into how AI specifically addresses the challenges faced by agile methodologies. These include things like rapid change, frequent iterations, and the need for real-time decision-making. The study also examines how AI can be used to increase knowledge sharing and drive innovation. For example, AI can provide real-time summaries and contextualize relevant lessons learned from previous projects, which in the long run improves collaboration, increases efficiency, and reduces the time spent searching for information. It also shortens the initiation phase for new projects, which was something that was raised as a challenge during the interviews.

Furthermore, through analysis of interviews with R&D teams at Vattenfall, this study identifies some opportunities for AI to automate documentation, flag potential risks early in the project, and provide real-time support for agile decision-making. AI can aggregate and analyze data from multiple sources to facilitate fast and informed decisions, allowing teams to remain focused on innovation. In addition, AI can play a crucial role in supporting knowledge sharing across the organization by capturing and distributing insights from previous and ongoing projects. This is valuable from a long-term strategic perspective, as it helps preserve knowledge within the company. This ensures that skills and learnings remain accessible and can benefit future projects. By investigating Vattenfall's R&D department, this study expands the understanding of the role of AI in optimizing agile project management and offers new insights into how AI can support innovation and streamline project workflows. This contributes to the literature on AI integration in agile methodologies, especially in terms of improving team collaboration, facilitating knowledge storage, creation and sharing.

Through the examination of AI applications and strategies, the study identifies challenges such as the resistance to AI adoption due to perceived loss of control over decision-making and opportunities like enhancing knowledge retention and sharing across distributed teams. The study also provides practical recommendations for integrating AI tools in agile workflows, particularly in iterative project cycles where immediate feedback and adjustments are critical. This investigation provides valuable insights into the potential of AI to improve efficiency, communication, and innovation in agile project management.

6.3 Optimizing Knowledge Sharing: Integrating AI with the SECI Model

The SECI model provides a framework for the dynamic exchange and application of knowledge (Adesina & Ocholla, 2019). Applying this model to the analysis of the study´s results, highlights that AI can enable each stage of this model, potentially leading to more effective collaboration, improved decision-making, and continuous learning across teams.

One of the key insights from the interviews was the value of informal knowledge sharing, particularly through meetings and direct communication. This practice aligns with the socialization phase of the SECI model, where tacit knowledge is shared through personal interactions (Rakan Aldmour, 2021). In agile methodologies, collaboration and experience sharing are essential to success (Kaur et al., 2015). AI can amplify socialization by facilitating communication and connecting team members based on their expertise and prior project experience. This enables the sharing of tacit knowledge, building on collective experience, and promoting a culture of continuous learning. For example, AI can match team members with similar challenges or skills gaps, enabling real-time discussions and knowledge exchange. In the implementation roadmap, this could be achieved through AI-powered knowledge hubs that automatically suggest team collaborations and foster spontaneous knowledge-sharing interactions. Interview participants also highlighted the value of AI-driven tools for connecting team members and enhancing knowledge flow across the organization.

Externalization refers to the process of transforming tacit knowledge into explicit knowledge that can be documented and shared (Rakan Aldmour, 2021). In agile teams, the ability to rapidly capture and distribute insights is essential to prevent knowledge loss (Kaur et al., 2015). By making internal knowledge accessible and usable across the team, externalization ensures that valuable insights are not confined to individuals. However, interviewees noted that documentation practices are often inconsistent, with key decisions and the rationale behind them not always being recorded. AI can significantly support this process by automatically summarizing meetings and structuring key takeaways, reducing the administrative burden while helping teams stay agile and ensuring that critical knowledge is retained and available for future use (Fairfax, 2024; Jin, 2024).

The combination part of the SECI model focuses on integrating and systematizing explicit knowledge across various projects (Rakan Aldmour, 2021). In agile environments, teams often work in parallel and need to share and apply insights from multiple projects (Kaur et al., 2015). During the interviews, participants highlighted the difficulties of efficiently accessing information from past projects, indicating a significant opportunity for AI to streamline the knowledge sharing process. AI can support this by analyzing past projects, identifying potential risks, and recognizing similarities between them. By making cross project learning more accessible, AI enables agile teams to optimize resources, uncover new connections, and enhance decision-making capabilities.

Following this, the internalization phase focuses on converting explicit knowledge back into tacit knowledge through practice and experience (Rakan Aldmour, 2021). Agile methodologies emphasize applying knowledge in real-time situations to foster learning and continuous improvement (Kaur et al., 2015). During the interviews, several participants highlighted the importance of learning from both successful and unsuccessful projects and applying these lessons in real-time scenarios. AI supports internalization by offering personalized, role-specific recommendations that guide teams in applying knowledge directly within their workflows (Fairfax, 2024; Jin, 2024). This helps improve decision-making, enhances agility, and strengthens the team's ability to adapt to changing project conditions.

The findings of this study demonstrate that AI has the potential to enhance the internalization phase of the SECI model by providing personalized, role-specific recommendations that facilitate real-time knowledge application. This ability to integrate AI into the agile process allows for improved decision-making and a more adaptive project environment. Regarding the first research question, AI not only supports the current knowledge management practices within agile projects at Vattenfall but also opens up possibilities for new forms of knowledge sharing. Specifically, AI could expand the traditional SECI model by creating a feedback loop that continuously evolves knowledge sharing, even as it happens. This could involve an AI-driven dynamic knowledge cycle, where AI not only suggests improvements and lessons learned but also captures tacit knowledge from team interactions, then transforms it into explicit knowledge for further application. By integrating AI, organizations can create an ongoing, self-improving knowledge ecosystem that continuously refines practices, adapts to new challenges, and fosters a deeper level of innovation.

6.4 Limitations and Future Work

There are several limitations to this study that should be considered. One key constraint is that the study primarily examined AI's role in knowledge management and innovation but did not explore the technical challenges associated with AI implementation in depth. Issues such as integration with existing systems, software compatibility, technical infrastructure, and economical factors were not thoroughly addressed, and these factors must be carefully examined before implementation.

Additionally, while the study explored how AI aligns with agile methodologies, agile practices can vary significantly between teams. The results may not fully capture the nuances of different agile work environments. Further research could examine AI's impact across a broader range of agile teams within the organization. A deeper investigation into agile practices themselves may also be needed, as some documentation challenges may stem from the adoption of the agile framework. While the employees stated that they follow agile methodologies, direct observations of their workflows were not conducted to verify whether they truly adhere to agile principles in practice. Moreover, although the organization refers to its approach as "Vattenfall Agile," there is limited clarity around what this specifically entails and why it has been adapted.

Another limitation is that the study focused specifically on Vattenfall R&D, meaning the findings may not be fully generalizable to other organizations or industries. The recommendations are based on the specific needs and challenges faced by Vattenfall R&D. Additionally, the data collected relied heavily on interviews with a limited number of participants. While these interviews provided valuable insights, the small sample size may not fully capture the diverse perspectives within the organization, potentially limiting the breadth of understanding.

The study also touched upon challenges related to AI adoption but did not deeply investigate the long term consequences or potential obstacles that may arise after implementation. This remains an important area for further research, as there is a gap in literature in understanding the extended impact of AI adoption over time.

Lastly, the employees state that they are positive to an AI tool but it would be valuable to investigate how employees actually adapt AI tools in their work practices. Understanding user behavior, resistance to AI adoption, and the factors that influence successful implementation could provide deeper insights into how to develop more user friendly and effective AI systems. By addressing these areas, future research could offer a more comprehensive understanding of the long term, practical, and organizational implications of AI implementation, ensuring its effective use in fostering innovation and enhancing knowledge management.

In addition to addressing the outlined limitations, exploring behavioral aspects of AI adoption could provide valuable insights. One particularly promising area is the intersection of AI and nudging, which is further elaborated in the following section.

6.4.1 Behavioral Dimensions of AI Adoption

Despite the growing presence of AI in various industries, the integration of human behavior and motivation into AI implementation remains a largely unexplored area, particularly within the context of agile project management.

One promising approach is the use of nudging, subtle design choices that influence user behavior without restricting options, as a means to align human behavior with desired outcomes (Lindström et al., 2019). By combining AI with nudging techniques, the two elements reinforce each other: nudging facilitates user acceptance and adoption of AI tools, while AI increases the precision and scalability of nudging strategies. Within agile project management, where user autonomy and adaptability are crucial, this creates an especially powerful synergy.

This combination enables the development of a tool that not only automates and streamlines project management but also subtly guides users toward improved decision-making and more structured documentation. Through effective UI and UX design, this approach ensures that information is both accessible and intuitively presented (Ugochukwu Francis Ikwuanusi et al., 2024). By integrating nudging principles in this way, knowledge sharing becomes a natural part of the user's workflow. AI can then encourage engagement without feeling intrusive, ultimately supporting better documentation habits, more informed decisions, and a seamless integration of AI into the agile process - where information is continuously collected and used to drive innovation, collaboration, and long-term learning.

Positive reinforcement (Lindström et al., 2019) can also be a powerful tool in digital systems. For example, when a project is archived correctly in line with documentation requirements, a celebratory message accompanied by confetti - such as "Well done!" - could appear. This simple gesture promotes a sense of community and motivation among employees, reinforcing positive behavior. By combining AI with psychological theories on behavior and motivation, organizations can design systems that not only improve efficiency but also foster meaningful user engagement in a way that feels both rewarding and intuitive.

In parallel with these research efforts, it is also essential to consider the financial aspects of AI integration. A cost estimation should also be conducted to better understand the financial implications of AI adoption. However, it’s important to recognize that AI implementation is a long term investment. While the full benefits may not be immediately apparent, over time AI can significantly enhance efficiency, free up resources for more innovative work, and make Vattenfall a more attractive workplace.

7. Conclusion

This study conducted a qualitative analysis of the current knowledge management practices within agile projects at Vattenfall, combining interviews with employees and observational insights, in order to identify opportunities for AI-supported improvements. The outcome is an implementation mapping a strategic, four-phase roadmap designed to guide the development and integration of AI within agile projects. The study also outlines the types of documentation needed to support each phase of AI implementation. In addition, it presents both a conceptual solution to provide an overarching understanding, and a visual prototype illustrating how an AI-driven platform equipped with such tools might look in practice.

This research makes several contributions to the existing literature, particularly in the underexplored area of AI integration within agile frameworks. First, it highlights the paradox between AI's reliance on structured data and agile methodology's emphasis on minimal documentation - revealing a key tension between the need for clarity in AI systems and the flexibility prized in agile environments. Second, it identifies a significant gap in current research: the limited implementation of AI in agile project management. Despite AI's growing potential, its role in supporting agile practices, fostering innovation, and enabling knowledge sharing remains largely overlooked. Third, the study explores the potential of combining AI with nudging techniques, suggesting a powerful synergy where AI subtly influences decision-making and behavior to enhance agile workflows. Lastly, the study examines the integration of AI with the SECI model, a well-established framework for knowledge management, as a way to strengthen knowledge sharing and collaboration across teams. By enhancing the SECI process, AI can support continuous learning and more informed decision-making in agile contexts.

Together, these contributions offer valuable insights and directions for future research, enriching the understanding of how AI can be strategically leveraged to support and advance agile practices.

The study shows that implementing AI at Vattenfall R&D, potentially will support their work in enhancing knowledge sharing between individuals and projects, fostering increased innovation in the short term. By reducing administrative workload and optimizing resource allocation, AI can free up valuable time, enabling employees to focus on driving innovation, fostering creativity, and developing new ideas and solutions in the long term. By integrating AI within agile frameworks, businesses can foster an environment where knowledge sharing is streamlined, and innovation is accelerated. AI should be seen as an enabler of agile practices rather than a replacement, with its success depending on phased implementation, continuous training, and a culture of collaboration.

Several critical factors were identified that must be considered when implementing AI, especially in organizations where confidentiality is a major constraint. In this regard, the study provides an implementation plan that fills some gaps in the existing literature. By leveraging AI's capabilities teams can redirect their efforts toward more strategic and creative aspects of their projects, ultimately driving efficiency and innovation.

While financial aspects must be taken into account, the interviewees expressed a positive attitude toward a solution similar to the one proposed. Based on the findings, it is clear that the need for AI support in project processes exists at the company.

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9. Appendices

The appendices related to the study are listed below.

9.1 Interview Guide 1

- What tools or methods do you use to plan and follow up on projects?
- What works particularly well in your way of working?
- What are the biggest challenges in your projects? (e.g. planning, resource management, communication, decision-making)
- Are there any parts of the project that you find inefficient or time-consuming?
- How do you manage risks and changes during the course of a project?
- Are there any tasks in the project that you think could be streamlined or automated?
- What kind of support would you wish for to make project work smoother?
- Is there anything in the project where you think AI could be helpful?

9.2 Interview Guide 2

Part 1: The Role of AI in Agile Projects

Current use of AI

- Do you currently use AI in your project work? If yes, in what ways?
- Have you experienced that AI has improved anything in innovation projects?

Part 2: Knowledge Sharing in Agile Projects

Current knowledge sharing processes

- Why is VF Portal not being used as intended?
- How do you currently share knowledge between projects and teams?
- How do you convert experiences and LL from projects into documented knowledge?
- Do you use formal processes, or does it happen more informally?
- Are there any requirements or guidelines?

Examples of successful knowledge sharing

- How do you ensure that documented knowledge is actually used in future projects?
- Can you give an example of a situation where knowledge sharing worked really well?
- What made it successful?
- How do you think AI could be used to support knowledge sharing?

Part 3: Documentation for Learning and Future Projects

Documentation at the start of a project

- Do you actively look for information from previous projects before starting a new one? Give examples?
- What type of data or insights would be most helpful to have available?

Documentation during the project

- What is important to document continuously throughout the project?
- Is there anything you often forget to document but later wish you had?

Documentation after the project ends

- How is this documentation used in subsequent projects?
- What is the most important thing to highlight in the lessons learned to make them useful in the future?

Stakeholder

- What makes someone a suitable stakeholder?

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